SGA 7-I: Monodromy Groups in Algebraic Geometry

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Directed by A. Grothendieck, with the collaboration of M. Raynaud and D. S. Rim.

Title page

Seminar on Algebraic Geometry at Bois-Marie, 1967–1969

Monodromy Groups in Algebraic Geometry

(SGA 7 I)

Directed by A. Grothendieck
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Introduction

Let S be a scheme and let $f : X \to S$ be a morphism of schemes. If f is proper and smooth, and if the prime number ℓ is invertible on S, the ℓ-adic cohomology groups

$$H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$$

of the geometric fibres of f form a local ℓ-adic system on S. For f assumed only proper, these groups are the fibres of an ℓ-adic sheaf $R^i f_* \mathbb{Z}_\ell$ on S. The theory of vanishing cycles relates the ramification of this sheaf on S to the singularities of f.

We shall consider only the case where S is a henselian trait, i.e. the spectrum of a henselian discrete valuation ring. This is in practice the essential case. I refer to (I 2.1) for a heuristic description of the theory. Let $S = \mathrm{Spec}(V)$, let $k(\bar{\eta})$ be an algebraic closure of the field of fractions $k(\eta)$ of V, and let $k(\bar{s})$ be the corresponding algebraic closure of the residue field $k(s)$; thus $k(\bar{s})$ is the residue field of the normalization $\bar{V}$ of V in $k(\bar{\eta})$. In the particular case where X is proper and flat over S, of relative dimension n, and where f has only one point $x \in X_s$ at which it is not smooth, one defines $\mathrm{Gal}(\bar{\eta}/\eta)$-modules Φ^i, purely local in nature in a neighbourhood of x, zero for $i \notin [0, n]$ (I 4.2), and one constructs a long exact sequence of $\mathrm{Gal}(\bar{\eta}/\eta)$-modules

$$\cdots \longrightarrow H^i(X_s, \mathbb{Z}_\ell) \xrightarrow{\mathrm{sp}} H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell) \longrightarrow \Phi^i \longrightarrow H^{i+1}(X_s, \mathbb{Z}_\ell) \longrightarrow \cdots .$$

The $\mathrm{Gal}(\bar{s}/s)$-module $H^i(X_s, \mathbb{Z}_\ell)$ is regarded as a $\mathrm{Gal}(\bar{\eta}/\eta)$-module with trivial inertia action; sp is the specialization map. Criteria are also given, at present only in characteristic 0, for the Φ^i with i small to vanish; for example, if X is moreover locally a complete intersection, then $\Phi^i = 0$ for $i \neq n$ in the situation of I 4.5.

The monodromy theorem asserts that the action of the inertia subgroup I of $\mathrm{Gal}(\bar{\eta}/\eta)$ is quasi-unipotent: for $T \in I$, there exist integers $N > 0$, $M > 0$ such that the endomorphism

$$(T^M - I)^N$$

of $H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell)$ is zero. Two proofs of this theorem are given. The first (I 1.2), arithmetic in nature, applies as soon as the cohomology groups considered have reasonable finiteness properties. The key point is that, when $k(s)$ is of finite type over its prime subfield, the action of I is quasi-unipotent for every ℓ-adic representation of $\mathrm{Gal}(\bar{\eta}/\eta)$. The second proof, more geometric, requires purity and resolution of singularities; it is at present valid only in characteristic 0 or for

H^1. It gives useful information about the exponent of nilpotence N ($N \leq i+1$ for H^i), and, as will be seen later, lends itself well, over $\mathbb{C}$, to comparison with transcendental theory.

Applied to H^1 of abelian varieties, i.e. to their Tate modules, these results make it possible to study reduction modulo p. Thus two proofs are given of the stable reduction theorem for abelian varieties: after ramification, the connected special fibre of the Neron model is an extension of an abelian variety by a torus. The first proof (I 6) takes up again the arithmetic method of the monodromy theorem. The second (IX 3.6) rests on a much finer analysis of the Neron model and of its polarization properties.

Grothendieck's Exposes I to V were not written up. They have been summarized in Expose I. The results stated there are proved succinctly, but essentially completely. Expose II applies the method of Lefschetz pencils and (I 5.3) to the study of the fundamental group. For S a surface over an algebraically closed field k, of characteristic exponent p, one shows that the profinite group $\pi_1^{(p')}(S, s)$, completed away from p, is of finite pro-(p') presentation. As explained above, Exposes III to V do not exist.

Expose VI contains, with some complements, Schlessinger's deformation theory. Care is taken there to account for infinitesimal automorphisms of the objects being classified, and not to pass too abruptly to the functor of isomorphism classes of objects. A new construction is also given of

$$\operatorname{Ext}^1(L_{X/S}, -),$$

where $L_{X/S}$ is the relative cotangent complex. This expose serves in the rest of the seminar especially through the study made there of deformations of ordinary quadratic singularities.

Exposes VII and VIII are devoted to the theory of biextensions. This theory plays an essential role in Expose IX, to express what happens to a polarization when one passes from an abelian variety to its Neron model. Expose IX contains the proof of the stable reduction theorem for abelian varieties and various applications. The sequel to this seminar, SGA 7 II, by P. Deligne and N. Katz, will appear later.

Bures-sur-Yvette, May 1972,
P. Deligne

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Expose I. Summary of the first exposes of A. Grothendieck

written up by P. Deligne

Contents

0. Preliminaries

0.0. Terminology

We recall the following definitions.

(0.0.1) A *trait* is the spectrum of a discrete valuation ring. Its generic and closed points are often denoted by η and s.

(0.0.2) A ring is *strictly local* if it is henselian local with separably closed residue field. A scheme is strictly local if it is the spectrum of such a ring.

(0.0.3) A geometric point of S in this expose is a morphism $\bar{x} : \mathrm{Spec}(k) \to S$ with image $x \in S$, where $k(x) \to k$ and k is a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, one often writes $\bar{\eta}$ for a generic geometric point of S. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S localized at s.

(0.0.4) The strict localization of X at the geometric point $\bar{x}$ is as in SGA 4 VIII 4.4.

(0.0.5) A local, henselian local, or strictly local S-scheme is said to be essentially of finite type over S if it is the spectrum of a local ring, respectively the henselization of such a local ring, respectively the strict localization, of a scheme of finite type over S.

(0.0.6) $F \mapsto F(n)$ denotes the Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, i.e. if there exist $N \geq 1$ and $M \geq 1$ such that $(T^N - 1)^M = 0$.

0.1. Reminders on ℓ-adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p, and let ℓ be a prime prime to p.

The ℓ-adic cohomology groups of X were defined in SGA 4 and SGA 5:

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i-th cohomology group of the etale site X_{et} with values in $\mathbb{Z}/\ell^n$;

b) $H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n)$;

c) $H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

The ℓ-adic cohomology groups with proper supports are defined similarly from the groups $H^i_c(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases one can prove that these groups $H^i(X)$ have reasonable finiteness properties: cohomology with proper support; X proper over k; X smooth over k (using duality and Poincare duality); when Hironaka resolution of singularities is available for schemes of finite type over k of dimension at most $\dim X$; and for $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are of finite type, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and, except perhaps in the last case, one has split short exact sequences

$$0 \to H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \to H^i(X, \mathbb{Z}/\ell^n) \to \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \to 0.$$

Moreover, the proof of the finiteness theorem gives each time a proof of generic constructibility: if k is the field of fractions of an integral noetherian scheme S and if X is the generic fibre of $f : X_1 \to S$ of finite type, there exists a nonempty open $U \subset S$ such that, over U, the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and have formation compatible with every base change.

0.2. Geometric cohomology

We no longer suppose k algebraically closed. Let $\bar{k}$ be an algebraic closure (or separable closure, which amounts to the same thing). We consider the geometric cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extending scalars from k to $\bar{k}$. By transport of structure, $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$-cohomology, in the cases (0.1.1) to (0.1.5), one obtains continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Reminders on traits

Let S be a henselian trait, with notation $\eta, s, V, \bar{V}, \bar{\eta}, \bar{s}$ as in (0.0.1) and (0.0.3). Let p be the characteristic exponent of $k(s)$. The map $\mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Gal}(\bar{s}/s)$ has kernel the inertia group I. The subfield of $k(\bar{\eta})$ defined by I is the maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro-p subgroup P. The quotient I/P is the tame inertia group, and canonically

$$I/P \simeq \varprojlim_{(n,p)=1} \mu_n(k(\bar{s})) \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})).$$

If $\chi_n : I/P \to \mu_n(k(\bar{s}))$ is the canonical map and if u is an element of valuation $1/n$, then for $\sigma \in I$,

$$\sigma(u) = \chi_n(\sigma)\, u \quad (\text{mod elements of valuation } > 1/n).$$

In summary,

$$\mathrm{Gal}(\bar{\eta}/\eta) \supset I \supset P, \quad \mathrm{Gal}(\bar{\eta}/\eta)/I \simeq \mathrm{Gal}(\bar{s}/s), \quad I/P \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})),$$

where P is a pro-p group (trivial if $p = 1$), and conjugation by $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s}))$.

0.4. Abhyankar's lemma

More generally, let S be strictly local regular, let D be a regular divisor in S, and let p' be the characteristic exponent of the generic point of D. Let $\bar{\eta}$ be a geometric generic point of S and let $\bar{s}$ be the geometric point localized at the closed point s which is deduced from it. By Abhyankar's lemma the fundamental group $\pi_1(S - D, \bar{\eta})$ has an analogous devissage:

$$\pi_1(S - D, \bar{\eta}) \supset I \supset P, \quad \pi_1(S - D, \bar{\eta})/I \simeq \mathrm{Gal}(\bar{s}/s), \quad I/P \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})),$$

where P has no quotient of order prime to p'. The action by inner automorphisms of $\pi_1(S - D, \bar{\eta})/I$ on I/P is the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s}))$.

0.5. Neron desingularization

The method of Neron desingularization gives the following result.

Lemma 0.5.1. Let $S \to S_1$ be a morphism of henselian traits, with $S = \mathrm{Spec}(V)$, $S_1 = \mathrm{Spec}(V_1)$, and let ϖ be a uniformizer of V_1. Suppose that ϖ is also a uniformizer of V, and that the extensions $k(s)/k(s_1)$ and $k(\eta)/k(\eta_1)$ are separable. Then V is the filtered inductive limit of henselian V_1-algebras, smooth and essentially of finite type over V_1.

To construct the desired V_1-algebras, one takes $V_1 \subset B \subset V$ with B of finite type, and applies Neron desingularization to $\mathrm{Spec}(B)$.

Let $S = \mathrm{Spec}(V)$ be a henselian trait. If S is of equal characteristic, with uniformizer ϖ, (0.5.1) applies to $S_1 = \mathbb{F}_p[\varpi]_{(\varpi)}$ after the usual separability check; hence S is a limit of spectra of henselian algebras, smooth and essentially of finite type over a trait of finite type. If S is complete of mixed characteristic, with perfect residue field k, then V is an Eisenstein extension of the Cohen-Witt ring $W(k)$; after a small perturbation of the coefficients one reduces to a trait whose residue field is a radical extension of a field of finite type over $\mathbb{F}_p$, and again (0.5.1) applies.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime prime to the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in $\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that no finite extension of $k(s)$ contains all roots of unity of order a power of ℓ. Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for $\sigma \in I_1$.

The preceding condition is automatically satisfied when $k(s)$ is of finite type over the prime field, or purely inseparable over such a field.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η, and let H be a cohomology space of $X_{\bar\eta}$, with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If the condition of Proposition 1.1 is satisfied, the action of any element of I on H is quasi-unipotent.

The limit arguments of 0.5 allow one to remove this hypothesis.

Variant 1.3. The condition of Proposition 1.1 is superfluous.

Let H' be a $\mathbb{Z}_\ell$-cohomology space of $X_{\bar\eta}$ which gives rise to H, and put $H_1 = H'/\mathrm{torsion}$. One has $H = H_1 \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. After replacing the trait by a finite extension, one may suppose that $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on $H_1/\ell H_1$, and, in mixed characteristic, that the trait is complete with perfect residue field. Using the presentation of the trait as a limit from 0.5, the representation comes from a constructible constant-twisted sheaf on $S_i - D_i$, and the devissages (0.3.1) and (0.4.1) give commutative diagrams

$$\begin{array}{ccccc} \mathrm{Gal}(\bar\eta/\eta) & \twoheadrightarrow & I & \twoheadrightarrow & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar\eta) & \twoheadrightarrow & I_i & \twoheadrightarrow & P_i \end{array} \qquad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}^{(p')}(1) \\ \downarrow & & \| \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}^{(p')}(1). \end{array}$$

Since $\mathrm{Ker}(\mathrm{GL}(H_1) \to \mathrm{GL}(H_1/\ell H_1))$ is a pro-ℓ group, P_i acts trivially, and one applies the following variant.

Variant 1.4. With the notation of 0.4, let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in

$\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies the condition of 1.1 and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

Let S be a strictly local trait, and keep the notation of 0.3; here $s = \bar{s}$. Let $f : X \to S$ be a scheme of finite type over S. The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar{\eta}}$, and the inertia action on the latter, from local information on X and on the morphism f.

In transcendental theory, if $f : X \to D$ is a proper morphism from an analytic space to a disk, the vanishing cycles appear as follows. After shrinking D, X is a topological fibration over the punctured disk D^*, so the fibres X_t for $t \in D^*$ are all isomorphic. One represents the special fibre X_0 as obtained from the general fibre by contraction of polyhedra:

$$\begin{tikzcd} X_t \arrow[rr, "\text{contraction}"] \arrow[dr] & & X_0 \arrow[dl] \\ & D & \end{tikzcd}$$

The method of vanishing cycles then studies the difference between the cohomology of X_t and that of X_0 by means of the Leray spectral sequence of this contraction.

The geometric theory below is very close to this transcendental theory: the general point $t \in D$ is replaced by $\bar{\eta}$, the geometric generic point of S. Conversely, the transcendental theory can be modelled on the geometric one by replacing t or $\bar{\eta}$ by the universal covering of D^*.

Proposition 2.3. Let $u : X' \to X$ be etale, x' a geometric point of X' over the geometric point x of X_s. The homomorphism

$$(R^i\Psi)_x \longrightarrow (R^i\Psi)_{x'}$$

induced by u is an isomorphism.

Corollary 2.4. The formation of the Ψ^i and Φ^i commutes with strict localization at a point x.

Let A be the coefficient ring. If I acts on the geometric generic fibre, one has the exact sequence of sheaves

$$0 \to A_{X_s} \to R\Psi A \to R\Phi A \to 0$$

in the derived sense; for f proper this gives the long exact sequence

$$\cdots \to H^i(X_s, A) \to H^i(X_{\bar{\eta}}, A) \to H^i(X_s, R\Phi A) \to H^{i+1}(X_s, A) \to \cdots .$$

3. Geometric proof of the monodromy theorem

Let S be strictly local, X a normal excellent scheme regular away from a divisor D, and suppose that, etale locally, the special fibre is a divisor with normal crossings

$$X_s = \sum_{i \in C} a_i D_i.$$

The proof uses purity to describe the prime-to-p coverings of the complement and then takes limits. If x is a point of the special fibre and if C is the set of components through x, set

$$K : \bigoplus_{i \in C} \mathbb{Z}(1) \xrightarrow{(a_i)} \mathbb{Z}(1),$$

concentrated in degrees $-1, 0$.

Theorem 3.3. Assuming purity, the groups of tame vanishing cycles are described by

$$R^*\Psi_x \simeq H^*(K \otimes A) \otimes A[t], \qquad \deg(t) = 2,$$

and the inertia action on $R^1\Psi_x$ is obtained from a transitive action on a finite set of cardinal prime to p, determined by the multiplicities a_i.

Corollary 3.4. Let N be the least common multiple of the multiplicities of the components of the special fibre. If at most q' components meet at a point, then for T in tame inertia

$$(T^N - 1)^{q'+1} = 0 \quad \text{on } R^q\Psi.$$

Theorem 3.5. Let C be a smooth projective curve over the field of fractions of a discrete valuation ring. The inertia group acts on

$$H^1(C_{\bar\eta}, \mathbb{Q}_\ell)$$

by quasi-unipotent automorphisms of level 2: for some N and all $T \in I$,

$$(T^N - 1)^2 = 0.$$

If A is an abelian variety, the same conclusion holds for $H^1(A_{\bar\eta}, \mathbb{Q}_\ell)$ and for the Tate module. More generally, for every scheme of finite type over η, inertia acts quasi-unipotently on $H^i(X_{\bar\eta}, \mathbb{Q}_\ell)$; in characteristic zero, for X proper and smooth one obtains $(T-1)^{i+1} = 0$ on H^i after passing to an open subgroup of inertia.

4. Vanishing criteria for sheaves of vanishing cycles

Let S be a strictly local trait and $f : X \to S$ of finite type. Put $n = \dim X_{\bar\eta}$. If X' is the schematic closure of X_η in X, then on $X_s - X'_s$ one has $\Phi^0 = A$ and the other Φ^i vanish; on X' the Φ^i and Ψ^i relative to X/S or X'/S agree, and $\Phi^0 = 0$. Since X' is flat over S, this often reduces the problem to the flat case.

Theorem 4.2. One has $\Phi^i = 0$ for $i > n$. More precisely, if x is a point of X_s whose closure has dimension k and $\bar{x}$ is a geometric point localized at x, then

$$\Phi^i_{\bar{x}} = 0 \qquad (i > n - k).$$

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension $< d$, the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar\eta})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

To prove in certain cases that the small Φ^i vanish, one uses local duality, directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities is known in the dimensions considered.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \to S$ be of finite type and let x be a closed point of X_s. Suppose that x is an isolated non-smooth point in its fibre and that X has relative geometric depth $\geq n$ at x, i.e. near x, X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x, with $n < N - k$. Then

$$\Phi^i_{\bar{x}} = 0 \qquad (i < n).$$

Corollary 4.6. For S a henselian trait of characteristic 0, X/S a relative complete intersection of relative dimension n, and $x \in X_s$ an isolated non-smooth point in its fibre, one has

$$\Phi^i_{\bar{x}} = 0 \qquad (i \neq n).$$

Corollary 4.7. Under the hypotheses of 4.6, $\Phi^n_{\bar{x}}$ is a flat A-module.

Variant 4.8. If in 4.5 one suppresses the isolatedness condition and lets Z be the non-smooth locus of f, then

$$\Phi^i = 0 \qquad (i < n - \dim Z).$$

Proposition 4.10. Under the general hypotheses of 4.5, suppose that x is an isolated non-smooth point in its fibre; that for every point y of $X_{\bar{\eta}}$ specializing to x, with closure of dimension $d(y)$, one has $\mathrm{prof}_{\mathrm{et},y}(X_{\bar{\eta}}) \geq n - d(y)$; and that $\mathrm{prof}_{\mathrm{et},x}(X_s) \geq n$. Then

$$\Phi^i_{\bar{x}} = 0 \qquad (i < n-1).$$

5. Action of monodromy on fundamental groups

In this section, which corresponds to Grothendieck's expose of 19 November 1968, we use the semistable reduction theorem for curves. Artin and Winters gave a direct proof; it can also be deduced from the analogous theorem for abelian varieties. We shall also use Schlessinger's theory as presented in Rim's Expose VI.

Let k be algebraically closed, let C be a projective curve over k, and let $x_0, \ldots, x_n$ be finitely many closed points of C. We say that $X = (C; x_0, \ldots, x_n)$ is stable if C is reduced and has only ordinary double points with distinct tangents, the x_i are distinct smooth points of C, and if C_j is an irreducible component of C and E is the set of points of its normalization lying over a singular point of C or one of the x_i, then E has at least 3 elements when C_j has genus 0 and at least 1 element when C_j has genus 1.

The last condition is equivalent to the absence of non-trivial infinitesimal automorphisms, and also to the ampleness of the invertible sheaf

$$\omega\Big(\sum x_i\Big),$$

where ω is the dualizing sheaf.

Proposition 5.1.1. Let $X = (C; x_0, \ldots, x_n)$ be stable over the generic point η of a trait S, with C_η smooth. There exists a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X \otimes_\eta \eta'.$$

Let X_η be a geometrically connected scheme of finite type over the field of fractions of a strictly local trait S. For a geometric point ξ of $X_{\bar{\eta}}$ one has an exact sequence

$$0 \to \pi_1(X_{\bar{\eta}}, \xi) \to \pi_1(X, \xi) \to \mathrm{Gal}(\bar{\eta}/\eta) \to 0, \tag{5.2.1}$$

and hence a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the maximal prime-to-p quotient gives

$$\mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Aut}_{\mathrm{ext}}(\pi_1^{(p')}(X_{\bar{\eta}})). \tag{5.2.2}$$

If X_η has a rational point x and $\bar{x}$ is the geometric point over it, the sequence splits canonically and gives

$$[x] : \mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Aut}(\pi_1^{(p')}(X_{\bar{\eta}}, \bar{x})). \tag{5.2.3}$$

Theorem 5.3. The image of wild inertia P under (5.2.2), or equivalently under (5.2.3), is finite.

The proof first reduces to the case where X_η is geometrically normal and integral. Let X' be the normalization and let $X'' = X' \times_X X'$. After a finite extension of $k(\eta)$, the connected components of X' may be assumed geometrically normal and integral with rational base points, and those of X'' geometrically connected with rational base points. Choosing prime-to-p path classes invariant under P, the Van Kampen formulas show that if a finite-index subgroup $P' \subset P$ acts trivially on the fundamental groups of the components, then P' acts trivially on $\pi_1^{(p')}(X_{\bar{\eta}})$.

For X_η normal and $U \subset X_\eta$ open, $\pi_1(U)$ maps onto $\pi_1(X_\eta)$. Hence one may suppose X_η smooth and affine. By cutting with generic hyperplane sections and using Bertini, one reduces to X_η smooth, geometrically connected, and of dimension one. After finite extension of $k(\eta)$ one may suppose that X_η is the complement, in a smooth projective curve C, of finitely many rational points; applying semistable reduction, one reduces to a stable curve over S. The continuation begins with the stable-curve calculation on the next source page.

Theorem 5.4

Let S be a strictly local scheme, let $f : \bar{X} \to S$ be a proper and flat morphism whose special fibre is a curve whose only points of non-smoothness are ordinary double points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Put

$$X = \bar{X} - Y.$$

Let x be a rational point of X, let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U. Then the image of

$$\pi_1(U, \xi)$$

in

$$\mathrm{Aut}(\pi_1^{(p')}(X_\xi, x_\xi))$$

is abelian and prime to p.

One must keep in mind that one cannot make $\pi_1(U, \xi)$ act on $\pi_1^{(p')}(X_\xi, x_\xi)$ except because the groups $\pi_1^{(p')}$ satisfy the specialization theorem.

The proof reduces first to the case where S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is a formal power series ring over a Cohen ring,

$$S = \mathrm{Spec}\, W[[t_1, \ldots, t_n]],$$

and U is the complement of a normal-crossings divisor with equation

$$t_1 \cdots t_m = 0.$$

In this universal case, by Abhyankar's lemma, $\pi_1(U, \xi)$ is abelian and prime to p, whence the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by considering a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem

by P. Deligne

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \to S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ deduced from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Neron model of $A_{\eta'}$ over S', let $\mathcal{A}_{S',s'}$ be its special fibre, and let $\mathcal{A}^0_{S',s'}$ be the neutral component of $\mathcal{A}_{S',s'}$. The stable reduction theorem is the following.

Theorem 6.1. For a suitable S', $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}^0_{S',s'}$ is an extension of an abelian variety by a torus.

It follows rather easily from 6.1 that one may choose S' so that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field; the same argument would work when $k(s)$ is radicial over such a field.

Let ℓ be a prime number prime to the residual characteristic exponent p, let $T_\ell(A)$ be the Tate module, let

$$V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell,$$

and let

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(T_\ell(A))$$

be the natural representation. A preliminary extension of scalars reduces us to the case where the inertia group I acts unipotently, by 1.2, hence through its largest pro-ℓ quotient $\mathbb{Z}_\ell(1)$, cf. 0.3. Let T be the image in $\mathrm{GL}(T_\ell(A))$ of a generator of $\mathbb{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T-1)^r \neq 0$.

Lemma 6.2. The map

$$\sigma \in I, \quad v \in V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces morphisms and an isomorphism

$$\begin{tikzcd} V_\ell(A)_I \otimes \mathbb{Q}_\ell(r) \arrow[r, "M"] \arrow[d] & V_\ell(A)^I \\ V_\ell(A)/\operatorname{Ker}(T-1)^r \otimes \mathbb{Q}_\ell(r) \arrow[r, "\sim"'] & \operatorname{Im}(T-1)^r \arrow[u] \end{tikzcd} \tag{6.2.1}$$

The modules $V_\ell(A)_I$ and $V_\ell(A)^I$ are $\mathrm{Gal}(\bar{s}/s)$-modules, and M is Galois-equivariant.

6.3. An ℓ-adic representation

$$\tau : \mathrm{Gal}(\bar{s}/s) \longrightarrow \mathrm{GL}(V)$$

is said to be of weight n ($n \in \mathbb{Z}$) if the following condition is satisfied: for a suitable model X of $k(s)$, with X an irreducible variety of finite type over the prime field and $k(s) = k(X)$, the representation τ factors through $\pi_1(X, \bar{s})$, and, for x a closed point of X, the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value

$$|k(x)|^{n/2},$$

where F_x denotes geometric Frobenius Φ_x^{-1}.

If V (respectively W) is of weight n (respectively m) and $n \neq m$, then

$$\mathrm{Hom}_{\mathrm{Gal}}(V, W) = 0.$$

6.4. It follows from the universal property of the Neron model that

$$V_\ell(A)^I = V_\ell(\mathcal{A}^0_{S,s}).$$

If a is the dimension of the largest abelian quotient of $\mathcal{A}^0_{S,s}$, and m the dimension of the multiplicative part of $\mathcal{A}^0_{S,s}$, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2.

6.5. Let A^*_η be the abelian variety dual to A_η. Recall that $V_\ell(A^*)$ and $V_\ell(A)$, hence also $V_\ell(A^*)^I$ and $V_\ell(A)_I$, are in duality with values in $\mathbb{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m for A, it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1.

6.6. The Galois module $\mathbb{Q}_\ell(r)$ has weight $-2r$. Since $\operatorname{Im}(T-1)^r \neq 0$, one deduces from 6.4, 6.5, and the isomorphism 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

hence $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. Then 6.2 implies that $\operatorname{Im}(T-1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0; these spaces have the same dimension $m_1 \leq m$. Hence

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}^0_{S,s} \geq 2(a+m) \geq 2a + m + m_1 = 2d = 2 \dim A_{S,s}.$$

Thus A_η has stable reduction, since $\dim \mathcal{A}^0_{S,s} = a + m$. At the same time one obtains $m_1 = m$, i.e.

$$\operatorname{Im}(T-1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}^0_{S,s})$$

is the V_ℓ of the multiplicative part of $\mathcal{A}^0_{S,s}$.

6.7. To treat the general case, we shall use 0.5, cf. 1.3. We reduce to the case where

a) the conditions of 0.5.2 or 0.5.3 are satisfied;

b) $\operatorname{Gal}(\bar{\eta}/\eta)$ acts trivially on A_ℓ;

c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses we shall prove that A_η has stable reduction. Hypothesis a) permits one to apply 0.5.1 and to write

$$S = \varprojlim S_i,$$

as in 1.3, whose notation we resume. Let s_i be the closed point of S_i. For i sufficiently large, the neutral component $\mathcal{A}^0_S$ of the Neron model $\mathcal{A}_S$ comes from a smooth group scheme with connected fibres $\mathcal{A}_i$ over S_i, and $(\mathcal{A}_i)_\ell$ is constant on $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar{\eta})$ acts on $T_\ell(A)$, and acts trivially on

$$T_\ell(A)/\ell T_\ell(A) \simeq A_\ell.$$

As in 1.3, one sees that P_i acts trivially. Then, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}^0_S) = T_\ell(A)^I = T_\ell(A^{I_i}),$$

and (6.2.1) is a diagram of $\operatorname{Gal}(\bar{s}_i/s_i)$-modules to which one can apply the arguments of 6.4–6.6.

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Expose II. Finiteness properties of the fundamental group

by Michele Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k, and let

$$\pi_1^{(p')}(X)$$

be the largest quotient "prime to p" of $\pi_1(X)$. We show in this expose that, if X is a surface, $\pi_1^{(p')}(X)$ is topologically of finite presentation, and that the same is true for arbitrary X if one admits resolution of singularities.

In the case of a surface, the method of proof, due to J.-P. Murre, consists in reducing to the case where one has a fibration over $\mathbb{P}^1$ having the properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k. Then one can find a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow \mathbb{P}^1,$$

such that f has a section σ and the following conditions are satisfied:

(i) f is smooth at the points of $\sigma(\mathbb{P}^1)$;

(ii) the fibres of f are geometrically integral curves having only ordinary singularities;

(iii) there exists a non-empty open U of $\mathbb{P}^1$ such that the fibres of f at the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \to \mathbb{P}^r$ such that one can find a Lefschetz pencil for that embedding. Let

$$D = \{H_t\}_{t\in\mathbb{P}^1}$$

be such a pencil of hyperplanes. Recall that it has the following properties: its axis Δ cuts S transversally, so that the closed subscheme $\Delta \cap S$ is reduced and consists of a finite set of closed points $x_1, \dots, x_n$ of S; there exists a non-empty open $U \subset \mathbb{P}^1$ such that, for every $t \in U$, H_t cuts S transversally; and, for every $t \in \mathbb{P}^1 - U$, H_t cuts S transversally except at one point, which is an ordinary quadratic singular point.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t; the application that, to x, associates $t \in \mathbb{P}^1$ is a rational map $a : S \dashrightarrow \mathbb{P}^1$ whose domain of definition is $S - (\Delta \cap S)$. Consider the blow-up

$$g : S' \longrightarrow S$$

of the closed subscheme $S \cap \Delta$ of S. The scheme S' is regular and the fibre S'_{x_i} over a point x_i is identified with the projective fibre defined by the tangent space to S at x_i, hence is isomorphic to $\mathbb{P}^1$. Moreover the rational map $f = ag$ is everywhere defined. Finally, to each point x_i one associates the section of f which, to a point $t \in \mathbb{P}^1$, makes correspond the tangent vector at x_i to $H_t \cap S$.

Let us verify (i), (ii), and (iii). For each $t \in \mathbb{P}^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic: the morphism $S'_t \to S_t$ is an isomorphism except perhaps at the points x_i, but S_t is regular at those points; moreover S'_t is a complete intersection in S', hence integral, and therefore isomorphic to S_t. Conditions (i) and (iii) are just conditions a) and b) of the Lefschetz pencil. Finally f is smooth at the points of σ, since f is flat and the fibres S'_t are geometrically regular at the points $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S, and let $Y' = Y \times_S S'$. Then one can choose f, g, U so that $Y' \times_{\mathbb{P}^1} U$ is a normal-crossings divisor relative to U.

Indeed, if there exists a Lefschetz pencil for an embedding $S \to \mathbb{P}^r$, the Lefschetz pencils form an open subset of the variety of lines in $\mathbb{P}^r$. The same holds for the set of Lefschetz pencils whose axis does not meet Y and for which there exists a non-empty open $U_1 \subset \mathbb{P}^1$ such that $Y \cap H_t$ is smooth for $t \in U_1$. It then suffices to construct f and g from a pencil satisfying these two conditions, and to replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is topologically of finite generation if there exists an integer n such that G is isomorphic to a quotient of the free pro-group on n generators, $F(n)$. Let $K = \mathrm{Ker}(F(n) \to G)$. One says that G is topologically of finite presentation if, in addition, one can find finitely many elements $k_1, \dots, k_r$ of K such that the smallest closed invariant subgroup of K containing $k_1, \dots, k_r$ is equal to K.

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k. Assume that schemes of finite type and of dimension $\leq \dim(X)$ over an algebraic closure of k are strongly desingularizable (SGA 5 I 3.1.5); this condition is satisfied when $\dim X = 2$, by [1], or when $k = 0$, by [2]. Then the fundamental group

$$\pi_1^{(p')}(X)$$

is topologically of finite presentation.

By SGA 1 IX 4.10, one may suppose k algebraically closed.

1) Reduction to the case where X is a regular surface. If $f : X' \to X$ is a morphism of finite type, set $X'' = X' \times_X X'$, and denote by $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ the sets of connected components of X' and X'', respectively. When f is a morphism of effective descent for etale coverings, the fundamental group $\pi_1(X)$ can be computed explicitly in terms of the groups $\pi_1(X'_a)$ and $\pi_1(X''_b)$. It follows from SGA 1 IX 5.2, 5.3 that, if the $\pi_1(X'_a)$ are topologically of finite generation, so is $\pi_1(X)$; and, if the $\pi_1(X'_a)$ are topologically of finite presentation and the $\pi_1(X''_b)$ topologically of finite generation, then $\pi_1(X)$ is topologically of finite presentation.

Since X is desingularizable, one can find a regular scheme X' and a proper birational morphism $f : X' \to X$; since f is a morphism of effective descent for etale coverings, it is enough to prove the theorem under the additional hypothesis that X is regular. Indeed one first deduces that $\pi_1(X)$ is topologically of finite generation; knowing that $\pi_1(X')$ is topologically of finite presentation and the $\pi_1(X''_b)$ topologically of finite generation, one deduces that $\pi_1(X)$ is topologically of finite presentation.

Let now $(U_i)_{i\in I}$ be a finite covering of X by affine open subsets, and let

$$f : \coprod_i U_i \longrightarrow X$$

be the canonical morphism. Since f is a morphism of effective descent for etale coverings, one sees that it is enough to prove the theorem for the U_i; one is therefore reduced to the case where X is affine and smooth.

Assume from now on that X is affine, smooth over k, and $\dim X > 2$. One may then apply the Lefschetz theorem for quasi-projective schemes: if Y is a sufficiently general hyperplane section of X, one has an isomorphism

$$\pi_1(Y) \simeq \pi_1(X).$$

It is therefore enough to prove the theorem for Y, which reduces us to the case $\dim X = 2$.

2) Case where X is a smooth surface over k. By the theorem of strong desingularization of surfaces, one can find a smooth, integral, projective surface S and an open immersion $i : X \to S$ such that the divisor $Y = S - X$ has normal crossings. Apply Corollary 2.2 and keep its notation.

It is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically of finite presentation. Since X and X' are regular and since

$$\operatorname{codim}\left(X' - \bigcup_{1\le i\le n}\{x_i\}, X\right) \ge 2,$$

every etale covering of X' induces on $X - \bigcup_{1\le i\le n}\{x_i\}$ an etale covering which extends uniquely to an etale covering of X (SGA 2 1.11). This shows that one has an isomorphism

$$\pi_1(X') \simeq \pi_1(X).$$

Consider the Cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X'_U \\ {\scriptstyle h}\uparrow & & \uparrow{\scriptstyle h_U} \\ \mathbb{P}^1 & \longleftarrow & U \end{array}$$

where $h = f|_{X'}$. Let L be the set of prime numbers distinct from p, and let $\bar\eta$ be a geometric point above the generic point η of $\mathbb{P}^1$. Let us verify that the hypotheses of SGA 1 XIII 4.8 are satisfied. The morphism h has a section σ, because f does. Condition a) of SGA 1 XIII 4.7 is proved as follows: the fibres of h being geometrically integral, the morphism h is 0-acyclic and locally 0-acyclic; it remains to show that, for every locally constant sheaf of sets F on X', the pair (F, h) is cohomologically proper in dimension ≤ 0. If ξ is a closed point of $\mathbb{P}^1$ and Z denotes the integral closure of $\operatorname{Spec}\mathcal{O}_{\mathbb{P}^1,\xi}$ in $\bar\eta$, the scheme $X'_Z = X' \times_{\mathbb{P}^1} Z$ satisfies condition (S_2) at the closed points of X' and is regular at every other point, hence is normal. Therefore an etale covering of X'_Z whose restriction to $X'_{\bar\eta}$ is connected is necessarily connected. This proves condition a). Since X'_U is the complement in S'_U of a normal-crossings divisor relative to U, h_U is locally 1-aspherical for L (SGA 4 XV 2.1), and, for every finite constant sheaf F of L-groups on X'_U, the pair (F, h_U) is cohomologically proper in dimension ≤ 1 (SGA 1 XIII 2.4); this is condition b). Since $\mathbb{P}^1$ is normal and $T = \mathbb{P}^1 - U$ has codimension 1, one has

$$\mathrm{prof}_T^{\mathrm{et}}(S) \geq 2,$$

which gives condition c). The condition

$$\mathrm{prof}_x^{\mathrm{top}}(X') \geq 3$$

holds at every closed point of X' by purity, and gives condition e). Finally, for every point $t \in \mathbb{P}^1$, h is smooth at $\sigma(t)$, hence is locally 1-aspherical for L at $\sigma(t)$, giving condition d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, one has an exact sequence

$$1 \longrightarrow \pi_1^{(p')}(X'_{\bar{\eta}}, a) \longrightarrow \pi'_1(X'_U, a) \longrightarrow \pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K denote the image of $\pi_1(U, a)$ in

$$\mathrm{Aut}(\pi_1^{(p')}(X'_{\bar{\eta}}, a)).$$

The group of coinvariants under K,

$$\pi_1^{(p')}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\pi_1^{(p')}(X'_{\bar{\eta}}, a)$ by the closed invariant subgroup generated by the elements of the form $x^{-1}q(x)$, where $x \in \pi_1^{(p')}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. 4.7.4, one has an isomorphism

$$\pi_1^{(p')}(X', a) \simeq \pi_1^{(p')}(X'_{\bar{\eta}}, a)_K.$$

Let

$$\mathbb{P}^1 - U = \coprod_{1 \leq i \leq n} \{t_i\},$$

and, for each i, let $\bar{T}_i$ be the strict localization of $\mathbb{P}^1$ at a geometric point $\bar{t}_i$ over t_i, and let s_i be the generic point of $\bar{T}_i$. An etale covering of U whose restriction to each s_i is trivial extends to an etale covering of $\mathbb{P}^1$, hence is trivial. Thus the assertion above amounts to saying that $\pi_1(U, a)$ is the closed invariant subgroup generated by the images of the Galois groups $\mathrm{Gal}(\bar{k}(s_i), k(s_i))$ in $\pi_1(U, a)$. By I 5.3, the image K_i of $\mathrm{Gal}(\bar{k}(s_i), k(s_i))$ in $\mathrm{Aut}(\pi_1^{(p')}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed invariant subgroup containing all the K_i, and since $\pi_1^{(p')}(X'_{\bar{\eta}}, a)$ is topologically of finite presentation (SGA 1 XIII 3.2), this proves that $\pi_1^{(p')}(X'_{\bar{\eta}}, a)_K$, and therefore also $\pi_1^{(p')}(X)$, to which it is isomorphic, is topologically of finite presentation.

Remark 2.3.2. Resolution of singularities is not necessary for proving Theorem 2.3.1 in the case where X is the open complement of a normal-crossings divisor in a projective scheme smooth over k.

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SGA 7-I, Exposé VI

Formal Deformation Theory

by D. S. Rim

1. Formal existence theorem

We recall the notion of cofibered category (SGA 1, IV).

Let C be a fixed category. A cofibered category A over C is, by definition, a category A together with a functor

$$P : A \longrightarrow C$$

such that:

Ax. 1. For any map f in C and any object a in A with $P(a)$ equal to the source of f, there exists an f-morphism

$$\alpha : a \longrightarrow b$$

which is cocartesian, i.e. for any f-morphism $\alpha' : a \to b'$ there exists a unique T-morphism $\tau : b \to b'$ in A such that

$$\alpha' = \tau\alpha,$$

where T is the target of f.

Ax. 2. A composite of cocartesian morphisms in A is again cocartesian.

The following properties of a cofibered category $P : A \to C$ follow immediately from the definitions.

Cancellation. Let $\alpha, \beta : a \to a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $\gamma : b \to a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

Factorization. Given cocartesian maps $\alpha : a \to a'$ and $\beta : a \to a''$, there exists a cocartesian map $\gamma : a' \to a''$ such that $\gamma\alpha = \beta$ if and only if there exists $f : P(a') \to P(a'')$ in C such that $fP(\alpha) = P(\beta)$. Furthermore γ is unique.

Let A be a cofibered category over C. For each object S in C we denote by $A(S)$ the subcategory of A whose objects are the objects a in A such that $P(a) = S$, and

$$\mathrm{Hom}_{A(S)}(a, a') = \{\alpha \in \mathrm{Hom}_A(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over C is a cofibered category A over C such that $A(S)$ is a groupoid for every object S in C. We recall that a groupoid is a category in which every morphism is an isomorphism. Thus, a cofibered groupoid over C is a cofibered category $P : A \to C$ such that every map in A is cocartesian. Furthermore, for every map $\alpha \in \mathrm{Fl}(A)$, α is an isomorphism in A if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, we fix a complete noetherian local ring Λ with residue field k, and we set $\mathcal{C}_\Lambda$ to be the category of artinian local Λ-algebras with the same residue field k, where the maps are local homomorphisms over Λ. From 1.2 on, a cofibered groupoid A over $\mathcal{C}_\Lambda$ is always assumed to have the property that $A(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of one and only one element for any two objects a, b in $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in A will be called simply a $\mathcal{C}_\Lambda$-map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. A map $a' \xrightarrow{\alpha} a$ in A will be called surjective, by abuse of language, if $P(\alpha) : P(a') \to P(a)$ is surjective in $\mathcal{C}_\Lambda$.

Example 1.1(a). Let $F : C \to$ (categories) be a functor. Define the category $\int F$ as follows: an object in $\int F$ is a pair (S, a) where $S \in \mathrm{ob}(C)$ and $a \in \mathrm{ob}\, F(S)$, and a map

$$(S, a) \longrightarrow (S', a')$$

is a pair (f, u') where $f : S \to S'$ is a map in C and $u' : F(f)a \to a'$ is a map in $F(S')$. Composition of two maps

$$(S, a) \xrightarrow{(f,u')} (S', a') \xrightarrow{(g,u'')} (S'', a'')$$

is given by

$$(gf, u'' \cdot F(g)u') : (S, a) \longrightarrow (S'', a'').$$

With respect to the functor $P : \int F \to C$ given by $(S, a) \mapsto S$, $\int F$ becomes a cofibered category over C. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(C)$, then $\int F$ is a cofibered groupoid over C. In particular, a functor $F : C \to$ (groups) yields a cofibered group, and a functor $F : C \to$ (Ens) yields a cofibered discrete groupoid. We also note that a natural transformation $\varphi : F \to G$, where $F, G : C \to$ (categories) are functors, induces a cocartesian functor $\varphi : \int F \to \int G$ over C.

Example 1.1(b). Let X, Y be schemes over Λ together with a fixed isomorphism $X \otimes_\Lambda k \xrightarrow{\sim} Y \otimes_\Lambda k$. Then the functor

$$\mathrm{Isom}_{X,Y} : \mathcal{C}_\Lambda \longrightarrow \text{(groupoids)}$$

given by

$$\mathrm{Isom}_{X,Y}(S) = \{\text{isomorphisms } X \otimes_\Lambda S \xrightarrow{\sim} Y \otimes_\Lambda S \text{ inducing the fixed one over } k\}$$

yields a cofibered groupoid $P : \int \mathrm{Isom}_{X,Y} \to \mathcal{C}_\Lambda$.

Example 1.1(c). Let X_0 be a fixed scheme over k, and let M_{X_0} be the category consisting of deformations of X_0 over $\mathcal{C}_\Lambda$ (see §4). Thus an object in M_{X_0} is a pair (R, X) where R is in $\mathcal{C}_\Lambda$ and X is a deformation of X_0 to R. Then the functor

$$P : M_{X_0} \longrightarrow \mathcal{C}_\Lambda, \qquad (R, X) \longmapsto R$$

defines a cofibered groupoid over $\mathcal{C}_\Lambda$.

Example 1.1(d). For any groupoid X we denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects in X. If $P : F \to C$ is a cofibered groupoid, we obtain a functor $[F] : C \to$ (Ens) given by $[F](S) = [F(S)]$, and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[F]$.

Definition 1.2. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called semi-homogeneous if the following properties hold.

(1) Every diagram of solid arrows in A where $a' \to a$ is surjective can be completed into a commutative diagram including the dotted arrow:

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d] & a' \arrow[d] \\ a_1 \arrow[r] & a. \end{tikzcd}$$

(2) Let

$$R \times_k k[\varepsilon] \rightrightarrows R$$

be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k, and let

$$a' \rightrightarrows a$$

be a pair of π_1-, π_2-maps in A. Then there exists a $\mathcal{C}_\Lambda$-map $\gamma : a' \to a''$ with $\alpha = \beta\gamma$ if and only if $(\pi_1)_*(a') = (\pi_2)_*(a'')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that, if A is semi-homogeneous, then so is $[A]$; in particular 1.2(2) entails that the canonical map

$$[A(R \times_k k[\varepsilon])] \longrightarrow [A(R)] \times [A(k[\varepsilon])]$$

is bijective.

Remark 1.3(b). Consider a commutative diagram in a cofibered category A over $\mathcal{C}_\Lambda$:

$$\begin{tikzcd} & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd}$$

Set $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If we set a'' to be the direct image under the map

$$(P(\nu_1), P(\nu_2)) : R' \longrightarrow R_1 \times_R R_2,$$

we obtain a commutative diagram. Thus 1.2(1) can be replaced by:

(1)' Every diagram in A above $R_1 \leftarrow R' \rightarrow R_2$ in $\mathcal{C}_\Lambda$, where

$$R' \longrightarrow R_1 \times_R R_2$$

is surjective, can be completed to a commutative diagram, where the maps μ_i $(i = 1, 2)$ are projections.

Remark 1.3(c). If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then $[A(k[\varepsilon])]$ is canonically provided with a vector-space structure over k, where $k[\varepsilon]$ is the ring of dual numbers over k. Indeed, let V be the category of finite-dimensional vector spaces over k. Then any functor

$$H : V \longrightarrow (\mathrm{Ens})$$

preserving the product is canonically factored through the forgetful functor $V \to (\mathrm{Ens})$. On the other hand, the functor

$$G : V \longrightarrow \mathcal{C}_\Lambda, \qquad W \longmapsto k[W] = k \oplus W, \quad w^2 = 0,$$

transforms products in V into fibre products over k in $\mathcal{C}_\Lambda$. If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then

$$[A(k[V] \times_k k[W])] \longrightarrow [A(k[V])] \times [A(k[W])]$$

is bijective by 1.3(a), and the functor $V \mapsto [A(k[V])]$ commutes with products, hence the result.

Definition 1.4. (1) Let $R \in \mathrm{ob}(\mathcal{C}_\Lambda)$. A small extension of R in $\mathcal{C}_\Lambda$ is a surjective map $R' \to R$ in $\mathcal{C}_\Lambda$ such that $(\mathrm{Ker}\, f)^2 = 0$ and $\mathrm{Ker}\, f \simeq k$. In other words, $R' \xrightarrow{f} R$ in $\mathcal{C}_\Lambda$ is a small extension of R if

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, and let a be an object in A. A small prolongation of a is an arrow $a' \to a$ in A such that $P(a') \to P(a)$ is a small extension in $\mathcal{C}_\Lambda$. An arrow $a' \to a$

in A will be called versal for small prolongations of a if, for every small prolongation $a_1 \to a$ in A, there exists a commutative diagram

$$\begin{tikzcd} a' \arrow[rr] \arrow[dr] & & a_1 \arrow[dl] \\ & a & \end{tikzcd}$$

Proposition 1.5. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$.

(1) Let $R \times_k k[\varepsilon] \rightrightarrows R$ be the projection maps in $\mathcal{C}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \to (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\mathcal{C}_\Lambda$, and consider a commutative diagram

$$\begin{tikzcd} R' \times_R R' \arrow[r] \arrow[d] & R' \arrow[d, "f"] \\ R' \arrow[r, "f"'] & R \end{tikzcd} \qquad \text{above} \qquad \begin{tikzcd} a'' \arrow[r] \arrow[d] & a' \arrow[d] \\ a' \arrow[r] & a. \end{tikzcd}$$

Define the canonical map

$$\mu : R' \times_R R' \longrightarrow k[\operatorname{Ker} f], \qquad \mu(r_1, r_2) = r_1(0) + (r_2 - r_1).$$

If there exists a map $a' \to (\mu)_*(a'')$, then there exists a map $\gamma : a' \to a$ such that $\alpha = a\gamma$.

Proof. (1) The "only if" part is clear. Now assume that there exists a map $\varphi : a \to (\pi_2)_* a'$. If we set

$$h : R \longrightarrow R \times_k k[\varepsilon], \qquad h = 1 \times P(\varphi),$$

then the composite map $R \xrightarrow{h} R \times_k k[\varepsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \xrightarrow{\gamma} a'$ such that $\pi_1\gamma = 1_a$, where $P(\gamma) = h$ and $P(a') = \pi_1$. On the other hand, $\pi_2 h = P(\varphi)$ and hence there exists a π_2-map $h^* a \to (\pi_2)_* a'$. This means that $(\pi_1)^* h^* a = (\pi_2)^* \varphi$, and therefore by 1.2(2) we have a $\mathcal{C}_\Lambda$-map $u : h^* a \to a'$ such that $\alpha u = a_1$. Define $\beta : a \to a'$ by $\beta = u\gamma$. Then $\alpha\beta = \alpha u \gamma = a_1 \gamma = 1_a$.

(2) The map $1 \times f : R' \times_R R' \to R' \times_k k[\operatorname{Ker} f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2(1 \times f) = \mu$, where $\pi_2 : R' \times_k k[\operatorname{Ker} f] \to k[\operatorname{Ker} f]$ is the projection map. We note that $k[\operatorname{Ker} f] \simeq k[\varepsilon]$ since $R' \to R$ is a small extension. Therefore, if there exists a map $a' \to (\mu)_* a'' = (\pi_2)_*(1 \times f)_* a''$, then by (1) above there exists a map $\gamma' : a' \to a''$ such that $\alpha'\gamma' = 1_{a'}$. Define $\gamma : a' \to a$ by $\gamma = \beta\gamma'$. We then have $\alpha\gamma = a$.

Corollary 1.6. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$ and consider a diagram

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd}$$

in A, where a_1 is a small prolongation. Assume that a is versal for small prolongations of e. Then there exists a map $a' \xrightarrow{\rho} a_1$ such that $\alpha = \alpha_1 \rho$ in A if and only if there exists a map $f : P(a') \to P(a_1)$ such that $P(\alpha) = P(\alpha_1) \cdot f$ in $\mathcal{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1) \cdot f$, where $f : P(a') \to P(a_1)$. Replacing a' by its direct image under f, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Set $R' = P(a_1)$ and $R = P(a)$. Since A is semi-homogeneous, we get a commutative diagram

$$\begin{tikzcd}
 & a'' \arrow[dl, "\beta"'] \arrow[dr, "\beta_1"] & \\
a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\
 & a &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & R' \times_R R' \arrow[dl] \arrow[dr] & \\
R' \arrow[dr] & & R' \arrow[dl] \\
 & R &
\end{tikzcd}$$

above the corresponding diagram of rings. Since a is versal for small prolongations of e, so is a', and therefore our statement follows from 1.5(2).

Definition 1.7. For each object R in $\mathcal{C}_\Lambda$ we put

$$\bar{\Omega}_R = M/(\mathfrak{m}_\Lambda R + M^2),$$

where $\mathfrak{m}_\Lambda$ and M are the maximal ideals of Λ and R respectively. It is a finite-dimensional vector space over k, and $R \mapsto \bar{\Omega}_R$ defines a functor

$$\bar{\Omega} : \mathcal{C}_\Lambda \longrightarrow V,$$

where V is the category of finite-dimensional vector spaces over k. If A is a cofibered category over $\mathcal{C}_\Lambda$, then the composite functor

$$A \xrightarrow{P} \mathcal{C}_\Lambda \xrightarrow{\bar{\Omega}} V$$

is, by abuse of notation, denoted by $\bar{\Omega}$ again.

Proposition 1.8. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$. Then, given an object a in A versal for small prolongations of e, there exists $a' \to a$ such that:

(i) a' is versal for small prolongations of a;

(ii) $P(a') \to P(a)$ is surjective, and the kernel is annihilated by the maximal ideal of $P(a)$;

(iii) $\bar{\Omega}_{a'} \to \bar{\Omega}_a$ is an isomorphism.

Proof. Set $R = P(a)$ and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0$$

where S is a formal power-series ring over Λ in a finite number of variables, and $J \subset M^2$, where M is the maximal ideal of S. Let Σ be the set of ideals J_1 in S such that (i) $J \supset J_1 \supset MJ$ and (ii) there exists a π_1-morphism $a_1 \to a$ in A, where $\pi_1 : S/J_1 \to R$ is the canonical map. The property 1.2(1) of A, together with the fact that J/MJ is a finite-dimensional vector space over k, entails that there exists a smallest ideal in the family Σ. Let it be J' and choose a π'-morphism $a' \to a$, where $\pi' : S/J' \to R$ is the canonical map. Note that $\bar{\Omega}_{a'} \to \bar{\Omega}_a$ is an isomorphism since $J' \subset J \subset M^2$. We claim that $a' \to a$ is versal for small prolongations of a. Indeed, let $a_1 \to a$ be a small prolongation. If we put $R_1 = P(a_1)$, we obtain an exact commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & J/MJ \arrow[r] \arrow[d, "h"'] & S/MJ \arrow[r] \arrow[d, "h_1"] & R \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & k \arrow[r] & R_1 \arrow[r, "\pi_1"'] & R \arrow[r] & 0.
\end{tikzcd}$$

If $h = 0$, then $\pi_1 : R_1 \to R$ admits a cross-section so that h_1 induces $f : S/J' \to R_1$. If $h \neq 0$, then $h_1 : S/MJ \to R_1$ is surjective and hence h_1 induces $f : S/J' \to R_1$ by the definition of the ideal J'. Therefore in either case we obtain a commutative diagram

$$\begin{tikzcd} S/J' \arrow[r, "f"] \arrow[dr, "\pi'"'] & R_1 \arrow[d, "\pi_1"] \\ & R. \end{tikzcd}$$

Since a is versal for small prolongations of e, it follows from 1.6 that we obtain a commutative diagram

$$\begin{tikzcd} a' \arrow[r, "\rho"] \arrow[dr] & a_1 \arrow[d] \\ & a. \end{tikzcd}$$

Let $P : A \to \mathcal{C}_\Lambda$ be a cofibered category (in groupoids). It is then clear that we obtain a cofibered category

$$\mathrm{pro}(P) : \mathrm{pro}(A) \longrightarrow \mathrm{pro}(\mathcal{C}_\Lambda)$$

(in groupoids). For the definition of the category of pro-objects $\mathrm{pro}(A)$, see A. Grothendieck, Séminaire Bourbaki 195.

$$\begin{tikzcd} A \arrow[r] \arrow[d, "P"'] & \mathrm{pro}(A) \arrow[d, "\mathrm{pro}(P)"] \\ \mathcal{C}_\Lambda \arrow[r] & \mathrm{pro}(\mathcal{C}_\Lambda). \end{tikzcd}$$

Now let $\widehat{\mathcal{C}_\Lambda}$ be the full subcategory of $\mathrm{pro}(\mathcal{C}_\Lambda)$ in which

$$\mathrm{ob}(\widehat{\mathcal{C}_\Lambda}) = \{(R_i)_{i \in \mathbb{N}},\ R_i \in \mathrm{ob}(\mathcal{C}_\Lambda),\ \varphi_i : R_i \to R_{i-1} \text{ is surjective for all } i,$$

and induces isomorphisms

$$\mathfrak{m}_i/(\mathfrak{m}_\Lambda + \mathfrak{m}_i^2) \xrightarrow{\sim} \mathfrak{m}_{i-1}/(\mathfrak{m}_\Lambda + \mathfrak{m}_{i-1}^2)$$

for all sufficiently large $i\}$. Here $\mathbb{N}$ stands for the ordered category of natural numbers, and $\mathfrak{m}_i$ is the maximal ideal of R_i. We set $\widehat{A}$ to be the full subcategory of $\mathrm{pro}(A)$ in which

$$\mathrm{ob}(\widehat{A}) = \{a \in \mathrm{ob}(\mathrm{pro}(A)) \mid \mathrm{pro}(P)(a) \in \mathrm{ob}(\widehat{\mathcal{C}_\Lambda})\}.$$

We then obtain a cofibered category

$$\widehat{P} : \widehat{A} \longrightarrow \widehat{\mathcal{C}_\Lambda}$$

(in groupoids) with a commutative diagram

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, "P"'] & \widehat{A} \arrow[d, "\widehat{P}"] \\ \mathcal{C}_\Lambda \arrow[r, hook] & \widehat{\mathcal{C}_\Lambda}. \end{tikzcd}$$

We note that the category $\widehat{\mathcal{C}_\Lambda}$ is canonically equivalent to the category of complete local Λ-algebras of formally finite type over Λ with fixed residue field k. Henceforth we shall make such identification.

Definition 1.9. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. An object ξ in $\widehat{A}$ is called a quasi-initial object of $\widehat{A}$ if for every object a in $\widehat{A}$ there exists a map $\xi \to a$ in $\widehat{A}$. A quasi-initial object ξ of $\widehat{A}$ is called minimal if the morphism $\xi : h_R \to [\widehat{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])],$$

where $R = P(\xi)$. An object ξ in $\widehat{A}$ is called a versal formal object of A if every surjective map $\alpha : a' \to a$ in A induces a surjective map

$$\mathrm{Hom}_{\widehat{A}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{A}}(\xi, a).$$

A versal formal object ξ of A is called minimal if the morphism $\xi : h_R \to [\widehat{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])],$$

where $R = P(\xi)$.

Proposition 1.10. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$.

(i) Every (minimally) versal formal object of A is a quasi-initial (minimal) object of $\widehat{A}$.

(ii) If ξ is an object in $\widehat{A}$ such that

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])]$$

is injective, where $P(\xi) = R$, then every endomorphism of ξ in $\widehat{A}$ is an automorphism. In particular a quasi-initial minimal object of $\widehat{A}$, if it exists, is unique up to non-canonical isomorphism.

Proof. (i) Let ξ be a versal formal object of A. We must show that, for any object a in $\widehat{A}$, we have a map $\xi \to a$. By the definition of $\widehat{A}$, there exists a sequence of surjective maps in A

$$\cdots \longrightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \longrightarrow \cdots \longrightarrow a_1 \xrightarrow{\alpha_1} a_0$$

such that $a = \lim a_n$. Assume that there exist maps $\varphi_i : \xi \to a_i$ such that $\alpha_i \varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is a versal formal object of A and $\alpha_n : a_n \to a_{n-1}$ is surjective in A, we obtain a map $\xi \xrightarrow{\varphi_n} a_n$ such that $\alpha_n \varphi_n = \varphi_{n-1}$. If we set $\varphi = \lim \varphi_n$, we have $\varphi : \xi \to a$.

(ii) Let $\xi \xrightarrow{\alpha} \xi$ be a map in $\widehat{A}$. Then, for every $\lambda \in \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon])$, we have

$$\lambda\xi = \lambda\alpha\xi = (\lambda \cdot P(\alpha))\xi,$$

and hence by the hypothesis on ξ we have $\lambda = \lambda \cdot P(\alpha)$ for all $\lambda \in \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon])$. Therefore $P(\alpha) : R \to R$ is surjective. However, every surjective endomorphism of a noetherian ring is an automorphism; hence $P(\alpha)$ is an automorphism of R, and therefore α is an automorphism of ξ.

Theorem 1.11 (Schlessinger). Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. Then A admits a minimally versal formal object if and only if:

(i) A is semi-homogeneous;

(ii) $[A(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let ξ in $\widehat{A}$ be a minimally versal formal object of A, and consider a diagram

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

in A where q is surjective. Since ξ is a quasi-initial object of $\widehat{A}$, there exists a map $\xi \to a_1$. Since q is surjective, the composite map $\beta = \beta_1\rho$ can be lifted to a', i.e. we obtain a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl] \arrow[d] \\ a_1 \arrow[r] & a' \arrow[d] \\ & a. \end{tikzcd}$$

This proves condition 1.2(1).

We now prove condition 1.2(2). Let $S \times_k k[\varepsilon] \rightrightarrows S$ be the projections, and let

$$\begin{tikzcd} a' \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & b' \end{tikzcd}$$

be the two maps in A. By the definition of ξ, we obtain a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl] \arrow[dr] & \\ a' \arrow[rr, shift left, "u"] \arrow[rr, shift right, "v"'] & & b'. \end{tikzcd}$$

In particular $\pi_1 P(u) = \pi_1 P(v)$. Assume now that $(\pi_2)^* a' = (\pi_2)^* b'$. Then $\pi_2 P(u) = \pi_2 P(v)$ because of the minimality of ξ, and hence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since A is a cofibered groupoid, we obtain a unique isomorphism $a' \xrightarrow{\sim} b'$ in $A(S \times_k k[\varepsilon])$ such that $\beta u = v$. Then $\alpha u = \alpha v$ entails that $\beta\alpha = \alpha$, since A is a cofibered category.

Sufficiency. Let $e_1, \dots, e_n$ be objects in $A(k[\varepsilon])$ which form a k-basis of the vector space $[A(k[\varepsilon])]$, and choose a_1 in $A(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$ such that $(\pi_i)_* a_1 = e_i$. Then

$$a_1(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R_1, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$; in particular $a_1 \to e$ is versal for small prolongations of e. Choose $a_2 \to a_1$ which is versal for small prolongations of a_1, and such that $\bar{\Omega}_{a_2} \to \bar{\Omega}_{a_1}$ is bijective. We then choose $a_3 \to a_2$, versal for small prolongations of a_2, such that $\bar{\Omega}_{a_3} \to \bar{\Omega}_{a_2}$ is bijective, and so on, and set

$$\xi = \lim_n a_n.$$

We claim that ξ is a minimally versal formal object of A. Indeed, since $\bar{\Omega}_{a_n} \to \bar{\Omega}_{a_{n-1}}$ is bijective for all n, $\xi(k[\varepsilon])$ is bijective. Now consider a diagram

$$\begin{tikzcd} & a' \arrow[d, "q"] \\ \xi \arrow[ur, dashed] \arrow[r, "h"'] & a \end{tikzcd}$$

where q is surjective in A. We must show that h can be lifted to a'. For this, one may assume, by induction, that a' is a small prolongation of a. Since $P(a)$ is an artinian ring, $h : \xi \to a$ is factored as a composite map $\xi \to a_n \to a$ for some a_n. Since A is semi-homogeneous, we get a diagram

$$\begin{tikzcd} a'' \arrow[r] \arrow[d, "q'"'] & a' \arrow[d, "q"] \\ a_n \arrow[r] & a \end{tikzcd}$$

where q' is also a small prolongation. We then obtain a commutative diagram

$$\begin{tikzcd} & a'' \arrow[d] \\ \xi \arrow[r] \arrow[ur, dashed] & a_n, \end{tikzcd}$$

and therefore we are done.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow e$$

be an infinite sequence of arrows in A such that a_i is versal for small prolongations of a_{i-1}, $P(a_i) \to P(a_{i-1})$ is surjective, and $\bar{\Omega}_{a_i} \to \bar{\Omega}_{a_{i-1}}$ is bijective for $i \gg 0$. Then the above proof shows that $\lim a_i$ is a versal formal object of A.

Remark 1.13. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. If A has the properties 1.11(i) and (ii), then so does $[A]$, and $\xi \in \mathrm{ob}(\widehat{A})$ is a minimally versal formal object of A if and only if it is so for $[A]$, by 1.10(ii).

Remark 1.14. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, admitting a minimally versal formal object ξ. Then an invariant of $R = P(\xi)$, which is a complete local Λ-algebra of formally finite type over Λ, is automatically an invariant of A over $\mathcal{C}_\Lambda$. Therefore, for example, we shall say that A is formally smooth over $\mathcal{C}_\Lambda$ if R is formally smooth over Λ, i.e. a formal power-series algebra over Λ, or we shall say that A is irreducible over $\mathcal{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible, etc. In the next section we introduce the notion of smoothness of a functor $A \to B$ over $\mathcal{C}_\Lambda$, where A, B are cofibered categories in groupoids over $\mathcal{C}_\Lambda$. It turns out (cf. 2.4) that "A is formally smooth over $\mathcal{C}_\Lambda$" is equivalent to "$P : A \to \mathcal{C}_\Lambda$ is a smooth functor", and therefore no confusion arises. We also define the dimension of A over $\mathcal{C}_\Lambda$ by

$$\dim_\Lambda A = \dim_k R.$$

For this we note the following fact. Let $\mathcal{C}_k$ be the category of artinian local k-algebras with residue field k. Then $\mathcal{C}_k \to \mathcal{C}_\Lambda$ is a fully faithful embedding whose left adjoint functor $\mathcal{C}_\Lambda \to \mathcal{C}_k$ is given by

$$R \longmapsto k \otimes_\Lambda R.$$

Let $A|\mathcal{C}_k$ be the pull-back of $P : A \to \mathcal{C}_\Lambda$ under $\mathcal{C}_k \to \mathcal{C}_\Lambda$. Then it is trivial to see that if ξ is a minimally versal formal object for A over $\mathcal{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is a minimally versal formal object for $A|\mathcal{C}_k$ over $\mathcal{C}_k$, where $\xi \to k \otimes_\Lambda \xi$ is a cocartesian arrow above $R \to k \otimes_\Lambda R$. It follows that

$$\dim_\Lambda A = \dim_k (A|\mathcal{C}_k).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory in order to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. We assume that:

(a) the natural map from Λ to K is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ.

The most interesting case is when $K = K'$ and K is a finite inseparable extension of k.

We denote by $\widehat{\mathcal{C}}_{\Lambda,K}$ the category whose objects are the complete noetherian local rings R provided with a structure of Λ-algebra and with a Λ-epimorphism $s : R \to K$.

We denote by $\mathcal{C}_{\Lambda,K}$ the subcategory of $\widehat{\mathcal{C}}_{\Lambda,K}$ consisting of those (R,s) such that $\mathrm{Ker}(s)$ is an R-module of finite length. The category $\mathcal{C}_{\Lambda,K}$ can be identified with a full subcategory of $\mathrm{pro}(\mathcal{C}_\Lambda)$. If K is artinian, so is any object in $\mathcal{C}_{\Lambda,K}$.

For any finite-dimensional vector space V over K', we denote by $D_V(K)$ the algebra of dual numbers over K defined by V; its elements are of the form $\kappa+v\varepsilon$ with $\kappa\in K$, $v\in V$ and $\varepsilon^2=0$, and

$$(\kappa+v\varepsilon)(\kappa'+v'\varepsilon)=\kappa\kappa'+(\kappa v'+\kappa' v)\varepsilon.$$

The map $e: D_V(K)\to K$, $e(\kappa+v\varepsilon)=\kappa$, turns $D_V(K)$ into an object of $\mathcal{C}_{\Lambda,K}$. The role played before by $k[\varepsilon]$ will be played here by $D_{K'}(K)$.

As explained in 1.8, any cofibered groupoid A over $\mathcal{C}_{\Lambda,K}$ extends in a natural way to a cofibered category $\widehat{A}_K$ over $\widehat{\mathcal{C}}_{\Lambda,K}$. A map in A or in $\widehat{A}_K$ will be called surjective if the corresponding homomorphism of algebras in $\mathcal{C}_{\Lambda,K}$ or in $\widehat{\mathcal{C}}_{\Lambda,K}$ is surjective.

Continuation of 1.15: residue-field extensions

We assume that $A(K)=\{e\}$.

Definition 1.16. A cofibered groupoid A over $\mathcal{C}_{\Lambda,K}$ is called *semi-homogeneous* if the following properties hold.

(I) Every diagram of solid arrows in A

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d, dashed] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

where $a'\to a$ is surjective, can be completed to a commutative diagram including the dotted arrows.

(II) For every R, the natural map

$$[A(R\times_K D_{K'}(K))]\longrightarrow [A(R)]\times[A(D_{K'}(K))]$$

is bijective, where the right side is the corresponding fibre product of isomorphism classes.

If A is semi-homogeneous, so is $[A]$.

1.17. Let us first restrict A to the full subcategory of $\mathcal{C}_{\Lambda,K}$ whose objects are the $D_V(K)$. Let

$$\Omega=\Omega^1_{K/\Lambda}\otimes_K K',$$

and let $d: K\to\Omega$ be the derivation. A morphism $f: D_V(K)\to D_W(K)$ can be written

$$f(k+v\varepsilon)=k+(u(v)+D\,dk)\varepsilon,$$

with $u\in\mathrm{Hom}_{K'}(V,W)$ and $D\in\mathrm{Hom}_{K'}(\Omega,W)$. If f is identified with the pair (u,D), the composition law is

$$(u,D)\circ(u',D')=(uu',\ D+uD').$$

Condition (II) entails the following condition:

$$V\longmapsto[A(D_V(K))] \tag{II'}$$

commutes with products as a functor from K'-vector spaces to sets.

Lemma 1.18. Assume that condition (II$'$) is satisfied by A. Then:

(1) $[A(D_{K'}(K))]$ carries a unique K'-vector-space structure such that, functorially in V,

$$[A(D_V(K))] \simeq V \otimes_{K'} [A(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \mathrm{Hom}_{K'}(\Omega, K') \longrightarrow [A(D_{K'}(K))]$$

such that, for any $D \in \mathrm{Hom}_{K'}(\Omega, K')$, the endomorphism induced by $(1, D) : D_{K'}(K) \to D_{K'}(K)$ is

$$x \longmapsto x + \gamma(D).$$

(3) More generally, for any $(u, D) : D_V(K) \to D_W(K)$, the corresponding map

$$[A(D_V(K))] \simeq V \otimes [A(D_{K'}(K))] \longrightarrow [A(D_W(K))] \simeq W \otimes [A(D_{K'}(K))]$$

is

$$x \longmapsto u(x) + \gamma(D).$$

Proof. Put

$$F(V) = [A(D_V(K))], \qquad E = F(K').$$

Condition (II$'$), together with the fact that K' is a vector-space object in the category of finite-dimensional K'-vector spaces, gives the vector-space structure on $F(K')$ and the functorial identification $F(V) \simeq V \otimes E$.

We write $F(u, D)$ for the map induced by (u, D). Since

$$(u, D) = (1, D) \circ (u, 0),$$

one has

$$F(u, D) = F(D) \circ (u \otimes 1_E).$$

The functoriality of F amounts to

$$F(0) = \mathrm{id}, \qquad F(D) \circ F(D') = F(D + D'), \qquad (u \otimes 1_E) \circ F(D) = F(uD) \circ (u \otimes 1_E).$$

For $V = K'^n$, using the maps $i : V \to V \oplus K'^n$ and $p : V \oplus K'^n \to K'^n$, The commutative diagram used in this reduction is

$$\begin{tikzcd}
V \otimes E \arrow[r] \arrow[d, "F(D)"'] & (V \oplus K'^n) \otimes E \arrow[r] \arrow[d, "F(i,D)"] & K'^n \otimes E \arrow[d, equal] \\
V \otimes E \arrow[r] & (V \oplus K'^n) \otimes E \arrow[r] & K'^n \otimes E.
\end{tikzcd}$$

one sees that $F(D)$ is a translation:

$$F(D)(y) = y + \alpha(D).$$

The identities above imply

$$\alpha(D + D') = \alpha(D) + \alpha(D'), \qquad \alpha(uD) = u\,\alpha(D),$$

and hence $\alpha(D)$ is defined by a unique linear map $\gamma : \mathrm{Hom}(\Omega, K') \to E$. This proves the assertions.

Definition 1.19. Let ξ be a formal object of A.

(1) The object ξ is called *versal* if every surjective map $a' \to a$ in A induces a surjective map

$$\mathrm{Hom}_A(\xi, a') \longrightarrow \mathrm{Hom}_A(\xi, a).$$

(2) A versal formal object ξ, with image (R, ξ), is called *minimal* if, for any two maps $\varphi, \psi : R \to D_V(K)$, whenever $\varphi(\xi)$ and $\psi(\xi)$ are isomorphic there exists a derivation $D : K \to V$ such that

$$(\psi, D) \circ \xi = \xi,$$

or equivalently $\psi = \varphi - D \circ d$.

Theorem 1.20. If A is semi-homogeneous and

$$\dim_{K'}[A(D_{K'}(K))] < \infty,$$

then A admits a minimally versal formal object, and any two such objects are isomorphic.

Proof. With the notation of Lemma 1.18, let H^* be a subspace of $[A(D_{K'}(K))]$ supplementary to the image of γ, and let H be the dual of H^*. Put

$$R_1 = D_H(K).$$

Choose $\xi_1 \in A(R_1)$ whose image in

$$[A(R_1)] = H^* \otimes [A(D_{K'}(K))] = \mathrm{Hom}(H^*, [A(D_{K'}(K))])$$

is the inclusion of H^* in $[A(D_{K'}(K))]$. Then

$$\mathrm{Hom}_{\mathcal{C}_{\Lambda,K}}(D_{K'}(K), D_V(K)) \longrightarrow [A(D_V(K))]$$

is surjective, and the minimality condition of Definition 1.19 is valid for (R_1, ξ_1).

Choose $S \in \widehat{\mathcal{C}}_{\Lambda,K}$, formally smooth over Λ, and an epimorphism

$$\pi : S \longrightarrow R_1$$

such that $\mathrm{Ker}(\pi) \cap \mathfrak{m}_S^2 = 0$. If $\mathfrak{m}$ is the maximal ideal of S, define inductively, for $n \geq 1$,

$$R_n = S/\mathfrak{m}^n, \qquad \xi_n \in A(R_n),$$

by requiring that:

(a) $\mathfrak{m}^n$ be the intersection of all ideals $I \supset \mathfrak{m}^n$ of S such that ξ_n extends to an object of $A(S/I)$;

(b) ξ_n extend ξ_{n-1}.

For

$$R = \varprojlim R_n, \qquad \xi = \varprojlim \xi_n,$$

one verifies that (R, ξ) is a minimally versal formal object. The construction also gives uniqueness up to isomorphism.

2. Pro-representable cofibered groupoids

Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, admitting a minimally versal formal object ξ. In general, ξ alone does not allow one to recover A, up to $\mathcal{C}_\Lambda$-equivalence.

Many cofibered groupoids which occur in practice satisfy a stronger property than semi-homogeneity. To study them, it is convenient first to rephrase the preceding construction in terms of smooth functors. The notation and terminology of §1 remain in force.

Definition 2.1. Let

$$\varphi : A \longrightarrow B$$

be a functor of cofibered groupoids over $\mathcal{C}_\Lambda$. We say that φ is *smooth* if every surjective map $\beta : b' \to \varphi(a)$ in B can be completed to a commutative diagram

$$\begin{tikzcd} \varphi(a') \arrow[r] \arrow[d] & b' \arrow[d, "\beta"] \\ \varphi(a) \arrow[r, equal] & \varphi(a) \end{tikzcd} \qquad (*)$$

for some $a' \to a$ in A. We say that φ is *minimally smooth* if it is smooth and the induced map

$$[A(k[\varepsilon])] \longrightarrow [B(k[\varepsilon])]$$

is bijective.

Remark 2.2.

(a) The condition that φ be smooth is equivalent to asking that, in the preceding diagram, $\varphi(a') \to b'$ may be chosen to be a $\mathcal{C}_\Lambda$-isomorphism; it is also equivalent to imposing the lifting condition only for small prolongations $b' \to \varphi(a)$.

(b) For every cofibered groupoid A, the natural functor

$$A \longrightarrow [A]$$

is minimally smooth.

(c) If $R \in \widehat{\mathcal{C}}_\Lambda$, put

$$h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S) \qquad (S \in \mathcal{C}_\Lambda).$$

This is a discrete cofibered groupoid over $\mathcal{C}_\Lambda$. A map $R \to S$ in $\widehat{\mathcal{C}}_\Lambda$ induces $h_S \to h_R$; the latter is smooth if and only if S is a formal power-series ring over R.

(d) Given $\xi \in A$ with $P(\xi) = R$, choose, for each $f : R \to S$, a cocartesian image $\xi(f)$. This defines a functor

$$\xi^* : h_R \longrightarrow A.$$

It is smooth precisely when ξ is a versal formal object; it is minimally smooth precisely when ξ is minimally versal.

Proposition 2.3. For functors of cofibered groupoids

$$A \xrightarrow{\varphi} B \xrightarrow{\psi} C$$

over $\mathcal{C}_\Lambda$:

(a) if φ and ψ are smooth, then $\psi\varphi$ is smooth;

(b) if ψ and $\psi\varphi$ are smooth, then φ is smooth;

(c) if $\psi\varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. If A has a versal formal object ξ, then $\xi^* : h_R \to A$ is smooth, where $R = P(\xi)$. Hence

$$A \text{ is formally smooth over } \mathcal{C}_\Lambda$$

is equivalent to saying that R is a formal power-series algebra over Λ, and also to saying that the structural functor $A \to \mathcal{C}_\Lambda$ is smooth.

Definition 2.5. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called *homogeneous* if every diagram

$$\begin{tikzcd} a' \arrow[r] \arrow[d] & a \arrow[d] \\ b' \arrow[r] & b \end{tikzcd}$$

where $a' \to a$ is surjective admits a fibre product $a' \times_a b'$ in A such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6.

(a) Homogeneity implies semi-homogeneity. Indeed, for any solid diagram in A above a diagram in $\mathcal{C}_\Lambda$, the fibre-product condition supplies the missing object and the missing arrows whenever the relevant map in $\mathcal{C}_\Lambda$ is surjective.

(b) The usual fibre product of categories $X \times_Z Y$ has objects (x, y) with $\Phi(x) = \Psi(y)$. The fibre product in the sense of 2-categories has objects $(x, y; u)$, where $u : \Phi(x) \simeq \Psi(y)$ is an isomorphism, and arrows are pairs which make the evident square commute:

$$\begin{tikzcd} \Phi(x) \arrow[r, "u"] \arrow[d] & \Psi(y) \arrow[d] \\ \Phi(x') \arrow[r, "u'"'] & \Psi(y'). \end{tikzcd}$$

Thus the notions of homogeneous and semi-homogeneous groupoids, and of smooth functors, may be expressed by requiring certain functors to such 2-categorical fibre products to be equivalences, or to be essentially surjective on isomorphism classes.

(c) If A is homogeneous, then for finite-dimensional k-vector spaces V, W, the functor

$$A(k[V \oplus W]) \longrightarrow A(k[V]) \times A(k[W])$$

is an equivalence. In particular $\mathrm{Aut}(1_V)$ and $[A(k[\varepsilon])]$ have canonical k-vector-space structures.

(d) For every $R \in \mathcal{C}_\Lambda$, the discrete groupoid h_R is homogeneous. If A is semi-homogeneous, then $[A]$ is semi-homogeneous; homogeneity of A does not, in general, imply homogeneity of $[A]$.

Proposition 2.7. Let A be a homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$. The following conditions are equivalent:

(i) $[A]$ is homogeneous, that is,

$$[A(R' \times_R S)] \longrightarrow [A(R')] \times_{[A(R)]} [A(S)]$$

is bijective for every small extension $R' \to R$.

(ii) For every small prolongation $a' \to a$ in A, the map

$$\mathrm{Aut}_{A(R')}(a') \longrightarrow \mathrm{Aut}_{A(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. The implication (i)⇒(ii) is obtained by applying homogeneity of $[A]$ to the two objects $a' \times_a a'$ and $a' \times_a a'_\tau$, where $\tau \in \mathrm{Aut}_A(a)$. The reverse implication follows by comparing two objects over $R' \times_R S$ whose images over R' and S are isomorphic; the assumed lifting of automorphisms permits the comparison isomorphisms to be modified so that the two squares commute simultaneously.

The comparison is represented schematically by the solid-arrow diagram

$$
\begin{array}{ccccccccc}
 & a_2'' & & & & & R' \times_R S & & \\
\swarrow & \downarrow & & \searrow & & \swarrow & & \searrow & \\
a_2' & a_1'' & b_1 \xrightarrow{\tau} b_2 & & & R' & & S & \\
 & \swarrow & & \searrow & & \searrow & & \swarrow & \\
 & a_1' & & a_1 & & & & R, &
\end{array}
$$

with the left part in A and the right diamond in $\mathcal{C}_\Lambda$.

Let $R' \xrightarrow{f} R$ be a small extension. There is a canonical map

$$\mathrm{Ker}(f) \longrightarrow R' \times_R R', \qquad (r_1, r_2) \longmapsto r_1(0) + (r_2 - r_1),$$

and the corresponding map

$$D_f : \mathrm{Ker}(f) \longrightarrow R' \times_R R'[\mathrm{Ker}(f)]$$

is an isomorphism. If $a_i \to a$ $(i = 1, 2)$ are two f-maps in A, we write

$$\tau(a_1, a_2) \in A(k[\mathrm{Ker}(f)])$$

for the object obtained by transporting $a_1 \times_a a_2$ through this identification. Then

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2),$$

compatibly with the projection onto the first factor.

Lemma 2.8. Let A be homogeneous over $\mathcal{C}_\Lambda$, and let $a_i \to a$ $(i = 1, 2)$ be f-maps in A.

(1) There exists $p : a_1 \to a_2$ in A, with $a_2 p = a_1$, if and only if there exists an arrow $a_1 \to \tau(a_1, a_2)$ in A.

(2) There exists such a p with $P(p) = \mathrm{id}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in } [A(k[\mathrm{Ker}(f)])].$$

Proof. This is the precise form of the preceding fibre-product description. Such a p is equivalent to a section of

$$a_1 \times \tau(a_1, a_2) \longrightarrow a_1,$$

and the condition $P(p) = \mathrm{id}$ is exactly the condition that the corresponding arrow to $k[\mathrm{Ker}(f)]$ factor through the zero section.

Proposition 2.9. Let

$$A \xrightarrow{\varphi} B \xrightarrow{\psi} C$$

be functors of cofibered groupoids over $\mathcal{C}_\Lambda$.

(a) Assume that B is semi-homogeneous, that ψ is homogeneous, and that

$$[A(k[\varepsilon])] \longrightarrow [C(k[\varepsilon])]$$

is injective. Then $\psi\varphi$ is smooth if and only if φ and ψ are both smooth.

(b) If $\varphi : A \to B$ is smooth and A is semi-homogeneous, then B is semi-homogeneous.

Proof. For (a), assume $\psi\varphi$ is smooth. Since smoothness descends as in Proposition 2.3, it remains to lift a small prolongation $b' \to \varphi(a)$. Smoothness of $\psi\varphi$ gives a lift in A after applying ψ, and the semi-homogeneity of B, together with the injectivity on tangent classes, removes the obstruction.

In the proof of smoothness one completes, by dotted arrows, the diagram

$$\begin{tikzcd}
\varphi(a'_{n+1}) \arrow[r, dashed] \arrow[d] \arrow[dr] & \varphi(a'_n) \arrow[d] & b'_{n+1} \arrow[r] \arrow[d, "\beta_{n+1}"] & b'_n \arrow[d, "\beta_n"] \\
\varphi(a_{n+1}) \arrow[r] & \varphi(a_n) & \varphi(a_{n+1}) \arrow[r] & \varphi(a_n).
\end{tikzcd}$$

After using the semi-homogeneity of A, the corresponding diagram in B is a fibre product; this gives the required lift.

Conversely, if φ and ψ are smooth, their composite is smooth. Part (b) is obtained by applying the lifting condition defining smoothness to the semi-homogeneous diagrams in A.

Corollary 2.10.

(a) Let $\varphi : A \to B$ be a smooth functor of cofibered groupoids over $\mathcal{C}_\Lambda$. Let ξ be a versal formal object of A, and let η be a minimally versal formal object of B. If B is homogeneous, then for any map

$$\beta : \eta \longrightarrow \varphi(\xi)$$

in $\widehat{B}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \longrightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let A be a homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, and let ξ, η be versal formal objects of A. If η is minimal, then for any map

$$\alpha : \eta \longrightarrow \xi$$

in $\widehat{A}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\alpha)$.

Proof. A map $\beta : \eta \to \varphi(\xi)$ induces the commutative square

$$\begin{tikzcd} h_{P(\xi)} \arrow[r, "\widetilde{\beta}"] \arrow[d, "\widetilde{\xi}"'] & h_{P(\eta)} \arrow[d, "\widetilde{\eta}"] \\ A \arrow[r, "\varphi"'] & B. \end{tikzcd}$$

Since $\widetilde{\xi}$ and φ are smooth, $\widetilde{\eta}\,\widetilde{\beta} = \varphi\widetilde{\xi}$ is smooth. Applying Proposition 2.9 and the minimality of η shows that $\widetilde{\beta}$ is smooth; equivalently $P(\xi)$ is a formal power-series ring over $P(\eta)$. Part (b) follows from (a) by taking $A = B$.

We now consider representability of a cofibered groupoid. Let C be a category. Each object $R \in C$ defines the functor

$$h_R : C \longrightarrow (\mathrm{Ens}), \qquad h_R(S) = \mathrm{Hom}_C(R, S),$$

and the Yoneda functor $h : C^\circ \to \mathrm{Hom}(C, \mathrm{Ens})$ is fully faithful.

A *cocategory* in C is a diagram

$$\begin{tikzcd} h_{R_1} \times_{h_{R_0}} h_{R_1} \arrow[r, "c"] & h_{R_1} \arrow[r, shift left, "s_1"] \arrow[r, shift right, "s_2"'] & h_{R_0} \arrow[l, bend left=60, "\varepsilon"] \end{tikzcd}$$

with $s_i\varepsilon = \mathrm{id}$ for $i = 1, 2$, such that c is associative and compatible with the s_i. Equivalently, the two composites

$$h_{R_1} \xrightarrow{\varepsilon} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\varepsilon\times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

and

$$h_{R_1} \xrightarrow{\varepsilon} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1\times\varepsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

are both identities.

If $(R_0, R_1, s_1, s_2, \varepsilon, c)$ is a cocategory in C, then for every $S \in C$ one obtains a category $h_{(R_0,R_1,\dots)}(S)$ by putting

$$\mathrm{Ob}\, h_{(R_0,R_1,\dots)}(S) = \mathrm{Hom}_C(R_0, S), \qquad \mathrm{Fl}\, h_{(R_0,R_1,\dots)}(S) = \mathrm{Hom}_C(R_1, S),$$

with composition induced by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

This defines a functor

$$h_{(R_0,R_1,\dots)} : C \longrightarrow (\text{categories}),$$

hence a cofibered category over C. A cogroupoid is a cocategory for which each $h_{(R_0,R_1,\dots)}(S)$ is a groupoid.

In practice, cocategories and cogroupoids arise as follows. Let $P : F \to C$ be a cofibered category and let ξ be a fixed object of F. Put $R_0 = P(\xi)$, let (R_0, C) be the category of arrows in C with source R_0, and choose a cocartesian section $\xi^\#$ of $F' \to (R_0, C)$ with $\xi^\#(\mathrm{id}_{R_0}) = \xi$. The functor

$$\mathrm{Hom}_\xi : (R_0 \amalg R_0, C) \longrightarrow (\mathrm{Ens})$$

is given by

$$X = (f_1, f_2) \longmapsto \mathrm{Hom}_{F(S)}(\xi^\#(f_2), \xi^\#(f_1)),$$

where S is the target of f_1 and f_2. If this functor is represented by

$$X_0 = (R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_1),$$

then R_1, together with the evident source, target, identity, and composition maps, defines a cocategory $(R_0, R_1, \ldots)$. The resulting functor

$$h_{(R_0,R_1,\ldots)} \longrightarrow F$$

is a C-equivalence if and only if ξ is quasi-initial in F over C.

Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\mathcal{C}}_\Lambda$ is called *smooth* if either of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1, and also via s_2;

(ii) the functor

$$h_{R_0} \longrightarrow h_{(R_0,R_1,\ldots)}$$

is smooth over $\mathcal{C}_\Lambda$.

It is called *minimally smooth*, or *normalized smooth*, if it is smooth and $h_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected, i.e.

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. The smoothness of $\varepsilon : h_{R_0} \to h_{(R_0,R_1,\ldots)}$ means precisely that, for every surjective map $(S', f') \to (S, f)$, a lifting of f to S', together with an isomorphism over S, can be lifted after replacing the isomorphism by one over S'. This is exactly the smoothness of $s_1 : h_{R_1} \to h_{R_0}$, and similarly for s_2.

Proposition 2.12. Let $(R_0, R_1, \ldots)$ be a cogroupoid in $\widehat{\mathcal{C}}_\Lambda$.

(a) If $(R_0, R_1, \ldots)$ is smooth, then $h_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following statements are equivalent:

  (i) $(R_0, R_1, \ldots)$ is smooth;

  (ii) $h_{(R_0,R_1,\ldots)}$ is homogeneous;

  (iii) $h_{(R_0,R_1,\ldots)}$ is semi-homogeneous.

(c) If $h_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $h_{(S_0,S_1,\ldots)}$ for some minimally smooth cogroupoid S_*.

Proof. For (a), consider a diagram

$$\begin{tikzcd}
(S_1, f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2, f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S, f) &
\end{tikzcd}$$

in $h_{(R_0,R_1,\ldots)}$, with $h_2 : S_2 \to S$ surjective. Since $s_1 : h_{R_1} \to h_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to some $v \in h_{R_1}(S_2)$ such that $s_1(v) = f_2$. This defines the object

$$(S_1 \times_S S_2,\ f_1 \times_S s_2(v))$$

and the commutative diagram

$$\begin{tikzcd}
& (S_1 \times_S S_2,\ f_1 \times s_2(v)) \arrow[dl, "\pi_1"'] \arrow[dr, "{(\pi_2,v^{-1})}"] & \\
(S_1, f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2, f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S, f) &
\end{tikzcd}$$

which is a fibre product. This proves homogeneity.

For (b), it remains to show (iii)⇒(i). If $h_{(R_0,R_1,\dots)}$ is semi-homogeneous, then $h_{(R_0,R_1,\dots)}$ admits a minimally versal formal object. On the other hand $(R_0, 1_{R_0})$ is a quasi-initial minimal object, and the equality on $k[\varepsilon]$-points makes it minimally versal. Hence

$$h_{R_0} \longrightarrow h_{(R_0,R_1,\dots)}$$

is minimally smooth, which is the required smoothness.

For (c), let (S_0, ξ_0) be a minimally versal formal object of $h_{(R_0,R_1,\dots)}$. For any object $\xi : R_0 \to T$, the functor $\mathrm{Isom}(\xi,\xi)$ on $\mathcal{C}_\Lambda/(T \times T)$ is represented by

$$R_1 \otimes_{R_0 \otimes_\Lambda R_0} (T \otimes_\Lambda T).$$

Passing to the inverse limit, $\mathrm{Isom}(\xi_0,\xi_0)$ is pro-represented by some S_1, and $h_{(R_0,R_1,\dots)}$ is equivalent to $h_{(S_0,S_1,\dots)}$. By (b), S_* is minimally smooth.

Definition 2.13. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called *pro-representable* (respectively *minimally pro-representable*) if it is $\mathcal{C}_\Lambda$-equivalent to

$$h_{(R_0,R_1,\dots)}$$

for some cogroupoid (respectively normalized cogroupoid) $(R_0, R_1, \dots)$ in $\widehat{\mathcal{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \dots)$ is a normalized cogroupoid in $\widehat{\mathcal{C}}_\Lambda$, then every homomorphism

$$(\sigma_0, \sigma_1) : (R_0, R_1, \dots) \longrightarrow (R_0, R_1, \dots)$$

which induces a $\mathcal{C}_\Lambda$-equivalence

$$h_{(\sigma_0,\sigma_1)} : h_{(R_0,R_1,\dots)} \longrightarrow h_{(R_0,R_1,\dots)}$$

is an isomorphism.

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2. Pro-representable groupoids, continued

Proof of Proposition 2.14. We keep the notation of the preceding statement. Suppose that a homomorphism (σ_0, σ_1) of a normalized cogroupoid induces a $\mathcal{C}_\Lambda$-equivalence on the associated cofibered groupoids. Applying the equivalence to the normalized base object gives a diagram of the form

$$\begin{tikzcd}
B_{R_0}(x,e) \arrow[r] \arrow[d] & h_{R_0}(x,e) \arrow[r] \arrow[d] & h_{(R_0,R_1,\dots)}(x,e) \arrow[d, "h_{(\sigma_0,\sigma_1)}"] \\
B_{R_0}(x,e) \arrow[r] & h_{R_0}(x,e) \arrow[r] & h_{(R_0,R_1,\dots)}(x,e).
\end{tikzcd}$$

Since, after passage to $k[\varepsilon]$, the relevant groupoid is totally disconnected, the induced tangent map is bijective. Hence the endomorphism $R_0 \to R_0$ is a surjective endomorphism of a complete local noetherian ring, and therefore an automorphism. The defining pull-back formula for R_1 then implies that $R_1 \to R_1$ is an automorphism as well. Thus (σ_0, σ_1) is an isomorphism.

Lemma 2.15. Let A, B be discrete groupoids over $\mathcal{C}_\Lambda$, and let $\varphi : A \to B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\mathcal{C}_\Lambda$-equivalence.

Proof. We must prove that $\varphi(S) : A(S) \to B(S)$ is bijective for every S. It is surjective by minimal smoothness, and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S. If $S' \twoheadrightarrow S$ is a small extension, homogeneity gives

$$S' \times_S S' \simeq S' \times_k k[\operatorname{Ker} f],$$

and therefore the commutative diagram

$$\begin{tikzcd}
A(S') \times_{A(S)} A(S') \arrow[r] \arrow[d, "\varphi\times\varphi"'] & A(S') \times A(k[\operatorname{Ker} f]) \arrow[d, "\varphi\times\varphi"] \\
B(S') \times_{B(S)} B(S') \arrow[r] & B(S') \times B(k[\operatorname{Ker} f]).
\end{tikzcd}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets under tangent groups identified by φ. If $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence bijective. □

Corollary 2.16. Let A be homogeneous over $\mathcal{C}_\Lambda$, with finite-dimensional tangent space. For every object ξ of A, the functor

$$\operatorname{Isom}_\xi : \mathcal{C}_\Lambda/R \longrightarrow \mathrm{Ens}, \qquad R = P(\xi),$$

is pro-representable.

Proof. The functor Isom_ξ is homogeneous: for a diagram $T_1 \to T \leftarrow T_2$ with one arrow surjective,

$$\operatorname{Isom}_\xi(T_1 \times_T T_2) \simeq \operatorname{Isom}_\xi(T_1) \times_{\operatorname{Isom}_\xi(T)} \operatorname{Isom}_\xi(T_2).$$

The formal existence theorem gives a minimally smooth functor from a pro-representable functor to Isom_ξ; Lemma 2.15 makes it an equivalence. □

Theorem 2.17. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is pro-representable by a normalized smooth cogroupoid if and only if

(1) A is homogeneous;

(2) $[A(k[\varepsilon])]$ and $\operatorname{Aut}(1_{k[\varepsilon]})$ are finite-dimensional vector spaces over k.

Proof. Smooth cogroupoids are homogeneous, and the tangent and automorphism spaces are kernels and cokernels of maps between spaces $\operatorname{Hom}(R_i, k[\varepsilon])$, hence finite-dimensional. Conversely, choose a minimally versal formal object ξ represented by R_0. Corollary 2.16 pro-represents the functor of isomorphisms between the two pull-backs of ξ over $R_0 \widehat{\otimes}_\Lambda R_0$ by an object R_1, carrying a universal isomorphism. Composition of isomorphisms gives a cogroupoid $(R_0, R_1, \ldots)$. The induced functor $h_{(R_0,R_1,\ldots)} \to A$ is minimally smooth, hence, by Lemma 2.15, an equivalence. □

Corollary 2.18. A functor $F : \mathcal{C}_\Lambda \to \mathrm{Ens}$ is pro-representable if and only if it is homogeneous and $\dim_k F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\widehat{\mathcal{C}}_\Lambda$ can be normalized, up to isomorphism.

For a cogroupoid $(R_0, R_1, \ldots)$, let $\overline{R}_1 = R_1/J$, where J is generated by $s_1(r) - s_2(r)$, $r \in R_0$. Then $(R_0, \overline{R}_1, \ldots)$ defines a formal group scheme over R_0.

Corollary 2.20. Let A be homogeneous and pro-represented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$. Then $(R_0, \overline{R}_1, \ldots)$ and $(k, k\widehat{\otimes}_{R_0}\overline{R}_1, \ldots)$ pro-represent the automorphism functors Aut_ξ and Aut^0. Moreover the following are equivalent:

(i) the formal group scheme is formally smooth over R_0;

(ii) for every surjection $a' \to a$ in A, the map $\mathrm{Aut}(a') \to \mathrm{Aut}(a)$ is surjective;

(iii) $\overline{R}_1$ pro-represents the functor of isomorphism classes over $\mathcal{C}_\Lambda$.

This is the combination of Proposition 2.7 with Lemma 2.15.

Remarks 2.21–2.22. Pulling a pro-representation back to $\mathcal{C}_\Lambda/k$ replaces the cogroupoid by its base change to k, preserving normalization and relative dimension. If A has two smooth pro-representations and one is normalized, then there is a comparison morphism of cogroupoids; the non-normalized base is a formal power-series extension of the normalized one, and the comparison becomes an isomorphism after this smooth base change.

3. The coherent sheaves D^i

We now recall the elementary obstruction theory of commutative algebra extensions in the language of Grothendieck cohomology on a site. Rings are commutative with unit. For an R-algebra A, let $\mathcal{C}_{A|R}$ be the category of arrows $B \to A$ in R-algebras, endowed with the topology whose coverings are surjective arrows. It has final object $A \to A$, initial object $R \to A$, fibre products and coproducts.

A sheaf F of A-modules on $\mathcal{C}_{A|R}$ is a functor such that, for every cover $B' \twoheadrightarrow B$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \times_B B')$$

is exact. Additive sheaves preserve coproducts. An A-module M gives the additive sheaf

$$\mathrm{Der}_R(-, M) : B \longmapsto \mathrm{Der}_R(B, M),$$

where M is viewed as a B-module through $B \to A$.

Let $\widetilde{\mathcal{C}}_{A|R}$ and $\widehat{\mathcal{C}}_{A|R}$ denote sheaves and presheaves. The inclusion of sheaves into presheaves has right derived functors $\mathcal{H}^q_{A|R}$, and

$$H^q(A|R, F) = [\mathcal{H}^q_{A|R}(F)](A).$$

An object $B \in \mathcal{C}_{A|R}$ and a ring homomorphism $S \to R$ give the functorial diagrams

$$\begin{array}{ccccccc} \widehat{\mathcal{C}}_{B|R} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} & \qquad & \widehat{\mathcal{C}}_{A|S} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ \widetilde{\mathcal{C}}_{B|R} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R} & & \widetilde{\mathcal{C}}_{A|S} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R}. \end{array}$$

For a covering $B' \to B$, the groups $H^q(\{B' \to B\}, F)$ are the cohomology groups of the Čech semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \rightrightarrows F(B' \times_B B' \times_B B') \rightrightarrows \cdots .$$

For a presheaf F, set

$$\mathcal{H}^q_{A|R}(F)(B) = \varinjlim_{\{B' \to B\}} H^q(\{B' \to B\}, F).$$

There is the usual spectral sequence

$$H^p(A|R, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(A|R, F),$$

and hence, in low degree,

$$\begin{aligned} &H^1(A|R,F) = \mathcal{H}^1(A|R,F), \\ &\qquad\qquad 0 \to \mathcal{H}^2(A|R,F) \to H^2(A|R,F) \to H^1(A|R,\mathcal{H}^1(F)) \\ &\qquad\qquad\quad \to \mathcal{H}^3(A|R,F). \end{aligned}$$

Lemma 3.2. If $P \twoheadrightarrow B$ is a cover by a polynomial R-algebra, then

$$H^n(\{P \to B\}, F) \xrightarrow{\sim} \mathcal{H}^n_{A|R}(F)(B).$$

Polynomial covers dominate all coverings, which proves the lemma.

Corollary 3.3. (a) If A is noetherian and F is of finite type, then $\mathcal{H}^n(F)$ is of finite type. (b) If P is polynomial over R, then $H^n(P|R,F) = 0$ for all $n > 0$. (c) Exactness of sheaves may be checked on polynomial R-algebras. Indeed, the polynomial subsite is cofinal and every surjection in it admits a section.

Lemma 3.5. For a diagram of R-algebras

$$\begin{tikzcd} B' \arrow[r] \arrow[d, two heads] & C \arrow[d] \\ B \arrow[r] & A \end{tikzcd}$$

and an R-algebra S:

(i) if $\mathrm{Tor}_1^R(B,S) = 0$, then

$$(B' \times_B C) \otimes_R S \simeq (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S);$$

(ii) if B' is R-flat and $\mathrm{Tor}_i^R(B,S) = \mathrm{Tor}_i^R(C,S) = 0$ for $0 < i \leq n$, then $\mathrm{Tor}_i^R(B' \times_B C, S) = 0$ for $0 < i \leq n$.

The proof compares the exact rows obtained from $0 \to J \to B' \to B \to 0$:

$$\begin{tikzcd} 0 \arrow[r] & J \otimes_R S \arrow[r] \arrow[d, equal] & (B' \times_B C) \otimes_R S \arrow[r] \arrow[d] & C \otimes_R S \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & J \otimes_R S \arrow[r] & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) \arrow[r] & C \otimes_R S \arrow[r] & 0. \end{tikzcd}$$

Corollary 3.6. If $\mathrm{Tor}_1^R(A,S) = 0$, then

$$H^n(A \otimes_R S|S, G) \simeq H^n(A|R, SG).$$

If S is R-flat, this gives $S\mathcal{H}^n(G) \simeq \mathcal{H}^n(SG)$. Polynomial covers remain Tor-acyclic under the hypotheses of Lemma 3.5.

Proposition 3.7. (a) If $\mathrm{Tor}_i^R(A,S) = 0$ for $0 < i \leq n$, then

$$\mathcal{H}^i(A \otimes_R S|S, F) \xrightarrow{\sim} \mathcal{H}^i(A|R, SF), \quad i \leq n.$$

(b) If P is polynomial over R, then for an additive sheaf F on $\mathcal{C}_{A \otimes_R P|R}$,

$$H^n(A \otimes_R P|R, F) \simeq H^n(A|R, F), \quad n > 0.$$

(c) For $S \to R \to A$ and additive F on $\mathcal{C}_{A|S}$, there is an exact sequence

$$0 \to H^0(A|R, F_R) \to H^0(A|S, F) \to H^0(R|S, F) \to H^1(A|R, F_R) \to \cdots ,$$

where $F_R(X) = \mathrm{Ker}(F(X) \to F(R))$. The proof compares the spectral sequences

$$\begin{tikzcd} H^p(A \otimes_R S|S, \mathcal{H}^q(F)) \arrow[r, Rightarrow] \arrow[d] & H^{p+q}(A \otimes_R S|S, F) \arrow[d] \\ H^p(A|R, \mathcal{H}^q(SF)) \arrow[r, Rightarrow] & H^{p+q}(A|R, SF) \end{tikzcd}$$

and then uses the Leray sequence for the forgetful functor.

Corollary 3.8. If $\mathrm{Tor}_i^R(A, B) = 0$ for $0 < i \le n$, then for additive F

$$H^i(A \otimes_R B|R, F) \simeq H^i(A|R, F) \oplus H^i(B|R, F), \quad i \le n.$$

This follows from the splitting diagram

$$\begin{tikzcd} H^i(A \otimes_R B|A, F_A) \arrow[r] \arrow[d, "\sim"'] & H^i(A \otimes_R B|R, F) \\ H^i(B|R, F_A) \arrow[r] & H^i(B|R, F). \end{tikzcd}$$

If moreover $A \otimes_R A \simeq A$ and $\mathrm{Tor}_i^R(A, A) = 0$, then $H^i(A|R, F) = 0$ for $i > 0$.

Corollary 3.9. For a multiplicative subset $S \subset A$, if F and F_S are additive, then

$$H^i(S^{-1}A|R, F) \simeq H^i(A|R, F).$$

For an A-module M, define

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)).$$

Choose $P \twoheadrightarrow A$ polynomial over R, with kernel I.

Lemma 3.10.

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)),$$
$$D^2(A|R, M) = H^1(A|R, \mathcal{H}^1(\mathrm{Der}_R(-, M))).$$

The proof uses the simplicial algebra $P^{(n)} = P \times_A \cdots \times_A P$ and the exact sequence

$$0 \to \mathrm{Der}_R(A, M) \to \mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M) \to H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \to 0.$$

Let $F \twoheadrightarrow I$ be a free P-module mapping onto I, and let $K_\bullet$ be the associated Koszul complex.

Lemma 3.11.

$$D^2(A|R, M) = \mathrm{Coker}(H^1(K_\bullet, M) \to \mathrm{Hom}_A(H_1(K_\bullet), M)).$$

This follows by replacing the cover $P \to A$ by the polynomial algebra $P[F]$; the obstruction term is

$$(F) \cap (F')/((F)(F')) \simeq \mathrm{Tor}_1^P(A, A) = H_1(K_\bullet).$$

Theorem 3.12. The groups D^i satisfy:

(1) a long exact sequence in $D^i(A|R,-)$ for every exact sequence of A-modules;

(2) for $S \to R \to A$, an exact sequence
$$0 \to D^0(A|R,M) \to D^0(A|S,M) \to D^0(R|S,M) \to D^1(A|R,M) \to \cdots;$$

(3) Tor-independent base change $D^i(A \otimes_R S|S,M) \simeq D^i(A|R,M)$;

(4) localization $D^i(S^{-1}A|R,S^{-1}M) \simeq S^{-1}D^i(A|R,M)$;

(5) the product formula $D^i(A \otimes_R B|R,M) \simeq D^i(A|R,M) \oplus D^i(B|R,M)$ under Tor-independence;

(6) compatibility with tensoring by a flat A-module, when A is essentially of finite type over a noetherian R;

(7) finiteness of $D^i(A|R,M)$ for M finite;

(8) the explicit formulas
$$D^0(A|R,M) = \mathrm{Der}_R(A,M),$$
$$D^1(A|R,M) = \mathrm{Coker}(\mathrm{Der}_R(P,M) \to \mathrm{Hom}_A(I/I^2,M)) = \mathrm{Exalcomm}(A|R,M),$$
$$D^2(A|R,M) = \mathrm{Coker}(H^1(K_\bullet,M) \to \mathrm{Hom}_A(H_1(K_\bullet),M)).$$

The extension interpretation of D^1 is given by the push-out diagram

$$\begin{tikzcd}
0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

Two maps $I \to M$ differing by a derivation of P give isomorphic extensions, and every extension arises this way.

Corollary 3.13.

(a) A is formally smooth over R iff $D^1(A|R,M) = 0$ for every M.

(b) If R is noetherian and A finite type over R, then A is a relative complete intersection iff $D^2(A|R,M) = 0$ for every M.

(c) Localization commutes with D^i.

(d) If A is local, essentially of finite type over noetherian R, and $\widehat{A}$ is its completion, then
$$D^i(\widehat{A}|R, \widehat{A} \otimes_A E) \simeq \widehat{A} \otimes_A D^i(A|R,E)$$
for every finite A-module E.

For (d), the final step compares algebra extensions and uses Artin–Rees. The completion comparison is expressed by the exact commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \widehat{E} \arrow[r] \arrow[d, equal] & B \arrow[r] \arrow[d] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \widehat{E} \arrow[r] & \widehat{B} \arrow[r] & \widehat{A} \arrow[r] & 0,
\end{tikzcd}$$

which proves the needed injectivity and surjectivity for D^1, and hence the assertion.

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Source scan pages 97–120: coherent deformation sheaves, formal moduli, and the beginning of formal Jacobian subschemes

3. The coherent sheaves D^i, conclusion

Corollary 3.14. Let R be noetherian and let A be an R-algebra of finite type, smooth everywhere except at one closed point $x \in \operatorname{Spec}(A)$. Then there are natural isomorphisms

$$D^1(A \mid R, A) \simeq D^1(A_x \mid R, A_x) \simeq D^1(\widehat{A}_x \mid R, \widehat{A}_x).$$

Indeed $D^1(A \mid R, A)$ is an A-module of finite type whose support is $\{x\}$. Hence localization and completion at x do not change it, and the assertion follows from 3.13.

Remark 3.15. Let R be a topological ring and A a topological R-algebra. For an A-module E set

$$D^i_{\text{top}}(A \mid R, E) = \varinjlim_{\alpha} D^i(A_\alpha \mid R_\alpha, A_\alpha \otimes_A E),$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α, I_α range through open ideals of A, respectively R. With this convention, saying that A is a formally smooth topological R-algebra means that

$$D^1_{\text{top}}(A \mid R, E) = 0$$

for every A-module E.

Let now X be a scheme over a ring R, and let E be a quasi-coherent $\mathcal{O}_X$-module. The sheaves

$$D^i_{X|R}(E) = D^i(X \mid R, E)$$

are defined on affine opens by

$$[D^i_{X|R}(E)](U) = D^i(\Gamma(U, \mathcal{O}_X) \mid R, \Gamma(U, E)).$$

If R is noetherian and X is of finite type over R, 3.13(a) gives this construction canonically as a quasi-coherent $\mathcal{O}_X$-module. When no ambiguity is possible we write

$$D^i_{X|R} = D^i_{X|R}(\mathcal{O}_X).$$

The preceding affine statements globalize to these sheaves. The following are the properties needed below.

Corollary 3.15. Let R be noetherian and X an R-scheme of finite type.

(a) For every coherent $\mathcal{O}_X$-module E, the sheaf $D^i_{X|R}(E)$ is coherent for all i, and

$$\operatorname{Supp} D^i_{X|R}(E) \subset X^{\text{Sing}} \quad (i > 0),$$

where X^{Sing} is the set of points at which X is not smooth over R.

(b) If X is flat over R, then for every R-algebra S there is the canonical base-change isomorphism

$$D^i_{X_S|S}(E_S) \simeq D^i_{X|R}(E)_S$$

for all i.

(c) If X is a relative complete intersection over R, then, for every surjection of rings $S \twoheadrightarrow R$, there is an exact sequence

$$0 \to D^1_{X|R}(E) \to D^1_{X|S}(E) \to \mathcal{H}om_{\mathcal{O}_X}(I/I^2 \otimes_R \mathcal{O}_X, E) \to 0,$$

where $I = \operatorname{Ker}(S \to R)$. If, moreover, R is local with residue field k and X is R-flat, then

$$0 \to D^1_{X_0|k}(\mathcal{O}_{X_0}) \to D^1_{X|S}(\mathcal{O}_X) \otimes_R k \to D^1(R \mid S, k) \otimes_k \mathcal{O}_{X_0} \to 0,$$

where $X_0 = X \otimes_R k$.

The proof is exactly the passage from the affine statements 3.12 and 3.13 to sheaves. In (c), relative complete-intersection hypotheses give $D^i_{X|R}(E) = 0$ for $i \geq 2$, and the exact sequence of 3.12(2), applied to $S \to R$, becomes the displayed exact sequence after identifying

$$D^1_{R|S}(E) = \mathcal{H}om_{\mathcal{O}_X}(I/I^2 \otimes_R \mathcal{O}_X, E).$$

The final sequence follows by reducing modulo the maximal ideal of R and using the base-change isomorphism.

Remark 3.16. The quasi-coherent sheaves $D^i_{X|R}(E)$ should occur as the $E_2^{0,i}$-terms of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory by homotopical algebra, generalizing methods of M. Andre and D. Quillen rather than only cohomological algebra. The present approach should also globalize by using Grothendieck cohomology on the category of schemes with a suitable topology extending the affine one.

4. Formal moduli of deformations

The cofibered categories of deformation problems are among the useful examples in which isomorphism classes of objects are often not pro-representable, but nevertheless admit formal versal objects. We return to the notation of paragraph 1. Thus Λ is a fixed complete local noetherian ring with residue field k, and $\mathcal{C}_\Lambda$ is the category of artinian local Λ-algebras with residue field k.

Let X_0 be a fixed scheme over k. An infinitesimal deformation of X_0 is a pair (R, X), where $R \in \mathrm{ob}(\mathcal{C}_\Lambda)$, X is a flat R-scheme, and there is a cartesian diagram

$$\begin{tikzcd} X_0 \arrow[r, "i_X"] \arrow[d] & X \arrow[d, "\text{flat}"] \\ \operatorname{Spec} k \arrow[r] & \operatorname{Spec} R. \end{tikzcd}$$

The induced morphism $X_0 \to X \times_{\operatorname{Spec} R} \operatorname{Spec} k$ is required to be an isomorphism. Since R is artinian local, i_X is a closed immersion and a homeomorphism on underlying spaces.

For two deformations (R, X) and (R', X'), an arrow $(R', X') \to (R, X)$ consists of a morphism $\varphi : R' \to R$ in $\mathcal{C}_\Lambda$ and a morphism $\xi : X \to X'$ fitting into the diagram

$$\begin{tikzcd} X \arrow[rr, bend left, "\xi"] \arrow[d] & X_0 \arrow[l, "i_X"'] \arrow[r, "i_{X'}"] \arrow[d] & X' \arrow[d] \\ \operatorname{Spec} R & \operatorname{Spec} k \arrow[l] \arrow[r] & \operatorname{Spec} R'. \end{tikzcd}$$

This gives a category $\mathcal{M}_{X_0}$ and a functor

$$p : \mathcal{M}_{X_0} \to \mathcal{C}_\Lambda, \qquad (R, X) \mapsto R.$$

If (R, X) is an object and $R \to S$ is a morphism in $\mathcal{C}_\Lambda$, then

$$X_S = X \times_{\operatorname{Spec} R} \operatorname{Spec} S$$

is a deformation of X_0 over S, and the cartesian square

$$\begin{tikzcd} X_S \arrow[r, "p_1"] \arrow[d, "\text{flat}"'] & X \arrow[d, "\text{flat}"] \\ \operatorname{Spec} S \arrow[r, "\operatorname{Spec}(\varphi)"'] & \operatorname{Spec} R \end{tikzcd}$$

shows that $\mathcal{M}_{X_0}$ is cofibered over $\mathcal{C}_\Lambda$.

4.1. The following elementary properties will be used repeatedly.

(a) Every arrow over the identity of R is an isomorphism. Thus $\mathcal{M}_{X_0}$ is cofibered in groupoids.

(b) If $X_0 = \operatorname{Spec} A_0$ is affine, then every deformation (R, X) is affine. The fiber $\mathcal{M}_{X_0}(R)$ is equivalent to the category whose objects are flat R-algebras A, together with an R-algebra map $j : A \to A_0$ inducing $A \otimes_R k \simeq A_0$.

(c) The corresponding formal category $\widehat{\mathcal{M}}_{X_0}$ has, over R, formal schemes $\mathfrak{X}$ adic over $\operatorname{Spf} R$, flat modulo every power of the maximal ideal. If X_0 is noetherian, this is the expected category of formal deformations of X_0.

(d) If $U_0 \subset X_0$ is open, restriction defines $\mathcal{M}_{X_0} \to \mathcal{M}_{U_0}$. Likewise a point $x_0 \in X_0$ gives functors to the deformation categories of $\operatorname{Spec} \mathcal{O}_{X_0,x_0}$ and $\operatorname{Spec} \widehat{\mathcal{O}}_{X_0,x_0}$.

Lemma 4.2. Consider a diagram of rings

$$\begin{tikzcd} R' \arrow[r] & R \\ R'' \arrow[u] \arrow[r, two heads] & R_0 \arrow[u] \end{tikzcd}$$

in which the indicated map is surjective. If E and E' are flat modules over the two upper rings and are identified after base change to R_0, then the fiber product module is flat over the fiber product ring. Equivalently, if

$$0 \to I \to R \to R' \to 0$$

is exact, the sequence

$$0 \to I \otimes E \to E \times E' \to E' \to 0$$

is exact and gives the Tor-vanishing needed for flatness.

Corollary 4.3. For a diagram

$$(R, X) \longleftarrow (R_0, X_0) \longrightarrow (R', X')$$

in $\mathcal{M}_{X_0}$, with the ring map $R' \to R_0$ surjective, the amalgamated sum $X \amalg_{X_0} X'$ is flat over $R \times_{R_0} R'$. Hence $\mathcal{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\mathcal{C}_\Lambda$.

Lemma 4.4. There is a canonical exact sequence

$$0 \to H^1(X_0, D^0_{X_0}) \to [\mathcal{M}_{X_0}(k[\varepsilon])] \to H^0(X_0, D^1_{X_0}).$$

If $X_0 = \operatorname{Spec} A_0$, the middle set is identified with isomorphism classes of k-algebra extensions of A_0 by A_0, hence with $D^1(A_0 \mid k, A_0)$. In general the map is obtained by restricting to affine opens. Its kernel is the set of locally trivial deformations, and the sheaf of germs of automorphisms of the trivial deformation $X_0 \times \operatorname{Spec} k[\varepsilon]$ is $D^0_{X_0}$; the usual Cech argument gives the displayed exact sequence.

Theorem 4.5. Let X_0 be a scheme of finite type over k. Suppose either that X_0 is proper over k, or that X_0 is affine and

$$X_0^{\mathrm{Sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite. Then:

(a) $\mathcal{M}_{X_0}$ admits a formal versal object over $\mathcal{C}_\Lambda$. If $(R, \mathfrak{X})$ is such a formal versal deformation and (S, Y) is an infinitesimal deformation, there is a morphism $R \to S$ and a cartesian diagram

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & \mathfrak{X} \arrow[d] \\ \operatorname{Spf} S \arrow[r] & \operatorname{Spf} R. \end{tikzcd}$$

Moreover $(R, \mathfrak{X})$ pro-represents the functor of isomorphism classes if and only if, for every surjection $R' \to R$ in $\mathcal{C}_\Lambda$ and every deformation (R', X'), the homomorphism

$$\operatorname{Aut}_{R'}(X' \mid X_0) \to \operatorname{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k, then $\mathcal{M}_{X_0}$ is smoothly pseudo-representable over $\mathcal{C}_\Lambda$.

The proof uses homogeneity from 4.3 and the tangent exact sequence 4.4. In the proper case, the coherent sheaves D^i have finite-dimensional cohomology. In the affine isolated-singularity case, $H^1(X_0, D^0_{X_0}) = 0$ and $H^0(X_0, D^1_{X_0})$ is finite-dimensional because the support is finite. The criterion for pro-representability is the automorphism criterion from 2.7.

Remarks 4.6. (a) If X_0 is affine, a formal versal deformation exists exactly when $D^1_{X_0}$ has finite support; this need not mean that the non-smooth locus itself is finite. (b) From now on, "deformation of X_0" means an object of $\mathcal{M}_{X_0}$; when ambiguity is possible, an object in the fiber over an artinian ring is called an infinitesimal deformation. (c) Although Λ is fixed, the number $\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2$ for a formal versal deformation depends only on X_0, not on the choice of Λ.

The same method applies to a pair $Y_0 \subset X_0$, where Y_0 is a closed subscheme. A deformation of the pair is a diagram

$$\begin{tikzcd} Y_0 \arrow[r] \arrow[d] & Y \arrow[d, "\text{flat}"] \\ X_0 \arrow[r] \arrow[d] & X \arrow[d, "\text{flat}"] \\ \operatorname{Spec} k \arrow[r] & \operatorname{Spec} R, \end{tikzcd}$$

which becomes the given pair after base change to k. These deformations form a homogeneous cofibered category $\mathcal{M}_{X_0,Y_0}$ over $\mathcal{C}_\Lambda$.

Lemma 4.7. Let I be the ideal sheaf of Y_0 in X_0. There is an exact sequence

$$0 \to H^0(Y_0, \mathcal{H}om_{\mathcal{O}_{Y_0}}(I/I^2, \mathcal{O}_{Y_0})) \to [\mathcal{M}_{X_0,Y_0}(k[\varepsilon])] \to [\mathcal{M}_{X_0}(k[\varepsilon])].$$

The proof identifies the kernel with deformations of Y_0 inside the trivial first-order deformation $X_0 \times \operatorname{Spec} k[\varepsilon]$. Such an object is equivalently a diagram

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & \mathcal{O}_{X_0} \arrow[r] \arrow[d] & \mathcal{O}_{Y_0} \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \mathcal{O}_{Y_0} \arrow[r, "\varepsilon"] & \mathcal{O}_Y \arrow[r] & \mathcal{O}_{Y_0} \arrow[r] & 0, \end{tikzcd}$$

and hence is determined by a morphism $I/I^2 \to \mathcal{O}_{Y_0}$.

Remark 4.8. A more general formula, implying Lemma 4.7, is given in SGA 5, III, 4.4. One can also define, for any S-scheme X, quasi-coherent sheaves $D^i_{X|S}$, extending the affine case, and one has a spectral sequence relating the global Ext-groups for the cotangent complex to the cohomology of these sheaves. In particular the low-degree terms recover the tangent, indeterminacy, and obstruction groups used in the preceding deformation arguments.

Theorem 4.9. Let X_0 be a k-scheme of finite type and let $Y_0 \subset X_0$ be a closed subscheme. Assume either that X_0 is proper over k, or that X_0 is affine, smooth outside a finite set of points, and Y_0 is proper over k. Then $\mathcal{M}_{X_0,Y_0}$ admits a formal versal deformation over $\mathcal{C}_\Lambda$, with the same pro-representability criterion by surjectivity on automorphism groups. If, moreover,

$$H^0(Y_0, \mathcal{H}om(I/I^2, \mathcal{O}_{Y_0}))$$

is finite-dimensional over k, then $\mathcal{M}_{X_0,Y_0}$ is smoothly pseudo-representable. This follows at once from 4.4, 4.6, 4.7 and the criteria 1.11, 2.7, and 2.13.

We now consider passage to localization and completion in the deformation theory of schemes. If $x_0 \in X_0$ is fixed, one has natural functors

$$\mathcal{M}_{X_0} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})},$$

and after completing formal deformations one obtains

$$\widehat{\mathcal{M}}_{X_0} \longrightarrow \widehat{\mathcal{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \widehat{\mathcal{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

If $(R, \mathfrak{X})$ is a formal deformation and

$$\mathfrak{X} = \varprojlim_{\nu} X_\nu, \qquad X_\nu = \mathfrak{X} \times_{\mathrm{Spf}\, R} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then the two images at x_0 are

$$\mathfrak{X}_{(x_0)} = \varprojlim_{\nu} \mathrm{Spec}(\mathcal{O}_{X_\nu,x_0}), \qquad \widehat{\mathfrak{X}}_{(x_0)} = \varprojlim_{\nu} \mathrm{Spec}(\widehat{\mathcal{O}}_{X_\nu,x_0}).$$

Theorem 4.10. Let $X_0 = \mathrm{Spec}(A_0)$ be an affine scheme of finite type over k.

(a) If X_0 is locally a complete intersection, then $\mathcal{M}_{X_0}$ is smooth over $\mathcal{C}_\Lambda$.

(b) Assume that, for a fixed closed point $x \in X_0$, the complement $X_0 - \{x\}$ is smooth over k. Then the functors

$$\mathcal{M}_{X_0} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}$$

are both minimally smooth. In particular a formal versal deformation of X_0 induces formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0,x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$.

Proof. For (a), it is enough to prove smoothness of the functor $[\mathcal{M}_{X_0}] \to \mathcal{C}_\Lambda$. Let $(R, \mathrm{Spec}\, A)$ be a deformation of $\mathrm{Spec}\, A_0$, and let $R' \to R$ be a small extension. The exact sequence for the composite $R' \to R \to A$, with coefficients in A_0, gives

$$\cdots \to D^1(A \mid R, A_0) \to D^1(A \mid R', A_0) \to D^1(R \mid R', k) \otimes_k A_0 \to D^2(A \mid R, A_0) \to \cdots .$$

Because A is R-flat,

$$D^i(A \mid R, A_0) = D^i(A_0 \mid k, A_0) \qquad (i > 0),$$

and the last group vanishes for $i = 2$ when A_0 is locally a complete intersection. Thus one finds an R'-algebra extension A' of A by A_0 mapping to the class of R', i.e. a lifting $(R', \mathrm{Spec}\, A')$ of $(R, \mathrm{Spec}\, A)$.

For (b), Corollary 3.14 gives canonical identifications of tangent spaces

$$D^1(A_0 \mid k, A_0)i \simeq D^1((A_0)_x \mid k, (A_0)_x) \simeq D^1(\widehat{(A_0)_x} \mid k, \widehat{(A_0)_x}).$$

Hence

$$[\mathcal{M}_{X_0}(k[\varepsilon])]i \simeq [\mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}(k[\varepsilon])]i \simeq [\mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}(k[\varepsilon])].$$

To prove minimal smoothness it remains, by 2.7(a), to show smoothness of the functors. Given $(R,$
$SpecA)$ and a small extension $R' \to R$, the same exact commutative diagram for $R' \to R \to A$, after localization and completion at x, shows that any local or completed local lifting may be lifted globally, because the groups $D^i(A \mid R, A_0)$ agree with their local and completed counterparts in positive degree.

Theorem 4.11. Let X_0 be a separated k-scheme of finite type, smooth outside the finite closed set $\{x_1, \dots, x_r\}$. Let $U_0^1, \dots, U_0^r$ be affine open neighbourhoods of these points such that $U_0^i \cap U_0^j$ is smooth over k for $i \neq j$. If

$$H^2(X_0, D^0_{X_0}) = 0,$$

then the natural functor

$$\mathcal{M}_{X_0} \to \mathcal{M}_{U_0^1} \times \cdots \times \mathcal{M}_{U_0^r}$$

is smooth. *Proof.* Complete $U_0^1, \dots, U_0^r$ to an affine open covering by choosing smooth affine opens U_0^j, $j > r$, such that $U_0^i \cap U_0^j$ is affine for all i, j. Let $R' \to R$ be a small extension. Suppose that a deformation (R, X) of X_0 and liftings (R', U'^i) of the prescribed deformations over U_0^i, $1 \leq i \leq r$, are given. For $j > r$, since U_0^j is smooth, 4.9(a) lifts $X|_{U_0^j}$ to a deformation (R', U'^j).

For each intersection $U_0^i \cap U_0^j$, smoothness gives isomorphisms

$$\tau_{ij} : U'^i|_{U_0^i \cap U_0^j} \xrightarrow{\sim} U'^j|_{U_0^i \cap U_0^j}$$

which reduce to the identity over R. The defect $\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$ is a Cech 2-cocycle with values in $D^0_{X_0}$. Its cohomology class lies in $H^2(X_0, D^0_{X_0})$; this class vanishes by hypothesis. Therefore the local liftings can be modified so that the cocycle condition holds, and then they glue to a deformation (R', X') with $X' \otimes_{R'} R = X$.

Remark 4.12. The preceding proof unnecessarily assumes that X_0 is separated. In fact, for a fixed deformation (R, X) and compatible deformations (R', U'^j) on an open covering, the deformations (R', V') of any open subset $V \subset X$, compatible with the already given data, form a gerbe in the sense of Giraud. Its band is $D^0_{X_0}$. Hence the obstruction to a section of this gerbe, i.e. to a global deformation compatible with the local data, is again in $H^2(X_0, D^0_{X_0})$.

Corollary 4.13. Let X_0 be a k-scheme of finite type, smooth outside a finite set

$$X_0^{\mathrm{Sing}} = \{x_1, \dots, x_r\},$$

and assume $H^2(X_0, D^0_{X_0}) = 0$. Then

$$\mathcal{M}_{X_0} \to \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_1})} \times \cdots \times \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_r})}$$

is smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0, and if $(R_i, \mathfrak{X}_i)$ are formal versal deformations of the completed local schemes, then

$$\dim \mathcal{M}_{X_0} = \dim H^1(X_0, D^0_{X_0}) + \sum_{i=1}^{r} \dim \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}.$$

If moreover X_0 is locally a complete intersection, then $\mathcal{M}_{X_0}$ is smooth. *Proof.* The first assertion is obtained from 4.10 and 4.9(b). Choose formal versal deformation rings R and R_i. The diagram

$$\begin{tikzcd} h_R \arrow[r] \arrow[d] & h_{R_1\widehat{\otimes}\cdots\widehat{\otimes}R_r} \arrow[d] \\ \mathcal{M}_{X_0} \arrow[r] & \prod_i \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})} \end{tikzcd}$$

shows, by 2.7(a), that $h_R \to h_{R_1\widehat{\otimes}\cdots\widehat{\otimes}R_r}$ is smooth after allowing $d = \dim_k H^1(X_0, D^0_{X_0})$ formal parameters. Taking tangent spaces gives the displayed dimension formula.

4.14. Explicit complete-intersection construction. Suppose

$$A_0 = k[x_1, \dots, x_n]/(f_1, \dots, f_r),$$

where $f_1, \dots, f_r$ is a regular sequence. The exact sequence

$$\cdots \to D^1(k[x_1, \dots, x_n] \mid k, A_0) \to \mathrm{Hom}_{A_0}(J/J^2, A_0) \to D^1(A_0 \mid k, A_0) \to 0$$

allows one to choose maps P_{uj} representing a k-basis of $D^1(A_0 \mid k, A_0)$. Put

$$R = \Lambda[[t_1, \dots, t_d]], \qquad B = R[[x_1, \dots, x_n]]/(G_1, \dots, G_r),$$

where $G_j = F_j - \sum_\nu t_\nu P_{uj}$, and where F_j is obtained from f_j by lifting its coefficients from k to Λ. Since the sequence remains regular modulo the maximal ideal, B is flat over R, and $k \otimes_R B = A_0$. The induced map

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \to D^1(A_0 \mid k, A_0)$$

is bijective by construction. Thus $(R, \mathrm{Spf}\, B)$ is a formal versal object. The same formula applies to the completed local complete-intersection algebra by 4.10(b).

Remarks 4.15–4.16. If X_0 is affine and locally a complete intersection with only isolated singularities, the construction shows that a formal versal deformation is the formalization of an R-scheme of finite type. If X_0 is not necessarily affine but $H^2(X_0, D^0_{X_0}) = 0$, the same statement globalizes.

The global theory of the relative cotangent complex gives a uniform formulation. Let $S_0 \subset S' \subset S$ be defined by ideals $J \supset I$ with $JI = 0$, let X' be flat over S', and put $L = L_{X_0/S_0}$. For the category F of flat schemes over S extending X', one has:

- (a) the automorphism group of an object is $\mathrm{Ext}^0(X_0, L, I_{X_0})$;
- (b) the set of isomorphism classes of objects, if non-empty, is a torsor under $\mathrm{Ext}^1(X_0, L, I_{X_0})$;
- (c) there is a canonical obstruction class in $\mathrm{Ext}^2(X_0, L, I_{X_0})$, whose vanishing is necessary and sufficient for non-emptiness.

Using

$$\mathrm{Ext}^i(X_0, L, I_{X_0}) = H^i(X_0, \mathrm{RHom}(L, I_{X_0})),$$

and the fact that the cohomology sheaves of $\mathrm{RHom}(L, I_{X_0})$ are the $D^q_{X_0}(I_{X_0})$, one obtains the spectral sequence

$$E_2^{p,q} = H^p(X_0, D^q_{X_0}(I_{X_0})) \Longrightarrow \mathrm{Ext}^{p+q}(X_0, L, I_{X_0}).$$

The first obstruction lies in $H^0(X_0, D^2)$; when it vanishes, the local solutions form a torsor under D^1, with obstruction in $H^1(X_0, D^1)$; after choosing a coherent system of local isomorphism classes, the remaining obstruction lies in a quotient of $H^2(X_0, D^0)$. This is the intrinsic form of the gluing argument used above.

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Since $\mathfrak{X}$ is an R-adic formal scheme rather than an ordinary scheme over R, we must say what is meant by its non-smooth locus.

We first recall Fitting ideals. Let A be a commutative ring and E an A-module of finite type. Choose a presentation

$$K \to F \to E \to 0,$$

where F is locally free of constant rank n. Define

$$I^p(E) = \mathrm{Im}\Big(\bigwedge^{n-p} K \otimes \bigwedge^{p} F^\vee \to A\Big) \qquad (p = 0, 1, 2, \ldots).$$

These ideals are independent of the chosen presentation and are the Fitting ideals of E. They satisfy:

(a) For every ring map $A \to B$, base change gives

$$B \otimes_A A/I^p(E) \simeq B/I^p(B \otimes_A E).$$

In particular localization commutes with I^p.

(b) The ideals are increasing, and $I^p(E)_x = A_x$ whenever $p > \dim_{k(x)} E(x)$. The support of $A/I^0(E)$ is $\mathrm{Supp}\, E$.

(c) Direct sums satisfy the usual convolution formula for Fitting ideals.

(d) If A is noetherian adic, $A = \varprojlim A_n$, and $E = \varprojlim E_n$, then

$$I^p(E) = \varprojlim I^p(E_n).$$

Thus for a scheme X and a quasi-coherent module E of finite presentation one obtains quasi-coherent ideals $I^p_X(E)$ by the affine formula

$$I^p_X(E)(U) = I^p(\Gamma(U, E))$$

on affine opens U. If $X \to Y$ is essentially of finite presentation, the relative Kahler differentials $\Omega^1_{X/Y}$ are of finite presentation, so the ideals $I^p_X(\Omega^1_{X/Y})$ define closed subschemes. This is the starting point for the formal Jacobian subschemes.

5. Formal Jacobian subschemes, continued

We use the notation introduced above for the Fitting ideals $I^p(E)$. For a morphism

$$f : X \longrightarrow Y$$

which is essentially of finite presentation, we denote by

$$I^p(X \mid Y) = I^p_X(\Omega^1_{X/Y})$$

the corresponding ideal sheaf on X, and by $J^p(X \mid Y)$ the closed subscheme which it defines. Thus $J^0(X \mid Y) \supset J^1(X \mid Y) \supset \cdots$.

Theorem 5.3. Let $f : X \to Y$ be essentially of finite presentation. Then:

(1) For every base change $Y' \to Y$, with $X' = X \times_Y Y'$, the canonical morphism

$$J^p(X' \mid Y') \longrightarrow J^p(X \mid Y) \times_Y Y'$$

is an isomorphism, for every p.

(2) For every $x \in X$, $y = f(x)$, the canonical local comparison is an isomorphism:

$$I^p_X(\Omega^1_{X/Y})_x \simeq I^p(\Omega^1_{\mathcal{O}_{X,x}/\mathcal{O}_{Y,y}}).$$

(3) The $J^p(X \mid Y)$ form a decreasing sequence of closed subschemes. A point x does not belong to $J^p(X \mid Y)$ once p is larger than

$$\dim_{\kappa(x)}(\Omega^1_{X/Y} \otimes \kappa(x)).$$

(4) One has $x \notin J^p(X \mid Y)$ if and only if, after replacing X by an open neighbourhood U of x, there exists a closed immersion

$$i : U \hookrightarrow U'$$

into a Y-scheme U' which is smooth over Y at $x' = i(x)$, and whose relative embedding dimension at x' is p.

(5) If X_1 and X_2 are two Y-schemes essentially of finite presentation and $X = X_1 \times_Y X_2$, then

$$I^p(X \mid Y) = \sum_{p_1+p_2=p} \mathrm{pr}_1^* I^{p_1}(X_1 \mid Y)\, \mathrm{pr}_2^* I^{p_2}(X_2 \mid Y).$$

Proof. Assertion (3) is the elementary monotonicity of Fitting ideals. Assertion (1) follows from the base-change formula for Fitting ideals and from

$$\Omega^1_{X'/Y'} \simeq \mathrm{pr}^* \Omega^1_{X/Y}.$$

Assertion (2) is the same statement after localization at x. Assertion (5) follows from the direct-sum formula for Fitting ideals applied to

$$\Omega^1_{X_1 \times_Y X_2/Y} \simeq \mathrm{pr}_1^* \Omega^1_{X_1/Y} \oplus \mathrm{pr}_2^* \Omega^1_{X_2/Y}.$$

For (4), suppose first that such an immersion $U \hookrightarrow U'$ exists. Since U' is smooth over Y at x', $\Omega^1_{U'/Y,x'}$ is free of rank p, and the exact conormal sequence

$$I/I^2 \longrightarrow \Omega^1_{U'/Y} \otimes \mathcal{O}_U \longrightarrow \Omega^1_{U/Y} \longrightarrow 0$$

shows that the p-th Fitting ideal is the unit ideal at x. Hence $x \notin J^p(U \mid Y)$, and therefore $x \notin J^p(X \mid Y)$.

Conversely, the question is local. Write $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and choose a presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \qquad P = A[x_1, \ldots, x_n].$$

The exact sequence

$$J/J^2 \longrightarrow B \otimes_P \Omega^1_{P/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

has middle term free of rank n. The condition $x \notin J^p(X \mid Y)$ means that a suitable $(n-p)$-minor of the Jacobian matrix of chosen generators of J is invertible at x. Thus there are elements

$$f_1, \ldots, f_{n-p} \in J$$

such that

$$U' = \operatorname{Spec}(P/(f_1, \dots, f_{n-p}))$$

is smooth over Y at the image x' of x, and $U = \operatorname{Spec} B$ embeds as a closed subscheme of U'. This gives the required local presentation.

The following result is the corresponding Jacobian criterion.

Theorem 5.4. Let $f : X \to Y$ be essentially of finite presentation. Suppose that there is an integer $n \geq 0$ such that either:

(1) f is flat and all fibers have dimension n;

(2) f is a relative complete intersection of virtual dimension n in the sense of SGA 6, VIII 1.9.

Then

$$x \notin J^n(X \mid Y) \iff f \text{ is smooth at } x.$$

Proof. By Theorem 5.3(4), $x \notin J^n(X \mid Y)$ if and only if locally X embeds into a Y-scheme U', smooth over Y at x', with relative dimension n. It remains to see that such an embedding is an isomorphism at x.

Let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be the corresponding exact sequence of local rings, with B' the local ring of U' at x' and B the local ring of X at x. Put $A = \mathcal{O}_{Y,y}$. In case (1), flatness gives the exact sequence on the closed fiber

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0,$$

where $\mathfrak{m}$ is the maximal ideal of A. The two fiber local rings have the same dimension n, and $B'/\mathfrak{m}B'$ is regular; hence $I/\mathfrak{m}I = 0$, whence $I = 0$.

In case (2), the conormal sequence

$$I/I^2 \longrightarrow B \otimes_{B'} \Omega^1_{B'/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

is exact. Since X is a relative complete intersection and U' is smooth, the first map is injective and the two modules which occur have the ranks prescribed by the virtual dimension. These ranks force $I/I^2 = 0$, hence $I = 0$.

Definition 5.5. Under the hypotheses of Theorem 5.4, one writes simply

$$I(X \mid Y) = I^n(X \mid Y), \qquad J(X \mid Y) = J^n(X \mid Y).$$

Thus $J(X \mid Y)$ is the Jacobian subscheme measuring failure of smoothness.

We now extend the construction to noetherian formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and E a coherent $\mathcal{O}_{\mathfrak{X}}$-module. Define

$$I^p_{\mathfrak{X}}(E)(U) = I^p(\Gamma(U, E))$$

for every affine open $U \subset \mathfrak{X}$. If $\mathfrak{X} = \varprojlim X_v$ and $E = \varprojlim E_v$, then

$$I^p_{\mathfrak{X}}(E) = \varprojlim_v I^p_{X_v}(E_v).$$

Let S be a noetherian formal scheme and let $\mathfrak{X}$ be an S-adic formal scheme, essentially of finite type. Let

$$\mathfrak{X} = \varprojlim X_v, \qquad S = \varprojlim S_v, \qquad X_v = \mathfrak{X} \times_S S_v.$$

Then

$$\Omega^1_{\mathfrak{X}/S} \simeq \varprojlim_v \Omega^1_{X_v/S_v}$$

is a coherent $\mathcal{O}_{\mathfrak{X}}$-module and is compatible with finite-type formal base change.

Definition 5.6. Define

$$I^p(\mathfrak{X} \mid S) = I^p_{\mathfrak{X}}(\Omega^1_{\mathfrak{X}/S})$$

and let $J^p(\mathfrak{X} \mid S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal. Equivalently

$$I^p(\mathfrak{X} \mid S) = \varprojlim_v I^p(X_v \mid S_v), \qquad J^p(\mathfrak{X} \mid S) = \varprojlim_v J^p(X_v \mid S_v).$$

There is the natural cartesian comparison with the closed fiber:

$$\begin{tikzcd} J^p(X_0 \mid S_0) \arrow[r] \arrow[d] & J^p(\mathfrak{X} \mid S) \arrow[d] \\ X_0 \arrow[r] & \mathfrak{X}. \end{tikzcd}$$

If x is an isolated point of $J^p(\mathfrak{X} \mid S)$, then the formal completion of $J^p(\mathfrak{X} \mid S)$ at x is canonically the corresponding connected component. In particular, if

$$|J^p(X_0 \mid S_0)| = \{x_1, \ldots, x_m\}$$

is finite and discrete, then

$$J^p(\mathfrak{X} \mid S) \simeq \coprod_{1 \le i \le m} J^p(\mathfrak{X} \mid S)_{(x_i)}.$$

Corollary 5.7. Let S be noetherian and formal, and let $\mathfrak{X}$ be an S-adic formal scheme essentially of finite type.

(1) The construction $J^p(\mathfrak{X} \mid S)$ commutes with formal base change:

$$J^p(\mathfrak{X}' \mid S') = J^p(\mathfrak{X} \mid S) \times_S S', \qquad \mathfrak{X}' = \mathfrak{X} \times_S S'.$$

(2) The connected components of $J^p(\mathfrak{X} \mid S)$ are in bijection with those of $J^p(X_0 \mid S_0)$, and for each isolated point x of $J^p(X_0 \mid S_0)$ there is a canonical isomorphism of completed local pieces

$$J^p(\mathfrak{X} \mid S)_{(x)} \simeq J^p(X_0 \mid S_0)_{(x)}.$$

In particular, in the finite discrete case one has the above disjoint-union decomposition.

(3) If S is an ordinary noetherian scheme, $S_0 \subset S$ a closed subscheme, $\widehat{S}$ the formal completion of S along S_0, and

$$\widehat{X} = X \times_S \widehat{S},$$

then

$$J^p(\widehat{X} \mid \widehat{S}) = J^p(X \mid S) \times_S \widehat{S}.$$

Thus the formal Jacobian subscheme is the completion of $J^p(X \mid S)$ along $J^p(X_0 \mid S_0)$.

(4) If a fixed integer n satisfies the hypotheses of Theorem 5.4 for all $X_v \to S_v$, then $x \notin J^n(\mathfrak{X} \mid S)$ if and only if $\mathfrak{X}$ is formally smooth over S at x.

Proof. The assertions follow directly from the corresponding statements for ordinary schemes, the compatibility of $\Omega^1_{\mathfrak{X}/S}$ with passage to X_v/S_v, and the base-change formula for Fitting ideals.

Remark 5.8. Let $\mathfrak{X}$ be S-adic, essentially of finite type, and let $x \in \mathfrak{X}$ with image $y \in S$. One can also consider the local formal $S_{(y)}$-scheme

$$\mathfrak{X}_{(x)} = \varprojlim_v \operatorname{Spec}(\mathcal{O}_{X_v,x_v}),$$

with the natural morphism $\mathfrak{X}_{(x)} \to \mathfrak{X}$. Although $\mathfrak{X}_{(x)}$ need not itself be essentially of finite type over $S_{(y)}$, the completed module of differentials is still coherent in the sense needed here:

$$\Omega^1_{\mathfrak{X}_{(x)}/S_{(y)}} \simeq \varprojlim_v \Omega^1_{\widehat{\mathcal{O}_{X_v,x_v}/\mathcal{O}_{S_v,y_v}}}.$$

This gives local formal subschemes $J^p(\mathfrak{X}_{(x)} \mid S)$, and if x is an isolated point of $J^p(\mathfrak{X} \mid S)$, the natural map

$$J^p(\mathfrak{X}_{(x)} \mid S) \longrightarrow J^p(\mathfrak{X} \mid S)$$

identifies the source with the completed local component at x.

We shall also need the image of the Jacobian subscheme in the base. The set-theoretic image may be described by kernels

$$\operatorname{Ker}\big(\mathcal{O}_S \to f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big),$$

but the Fitting-ideal formulation is more useful.

Definition 5.9. Let S be noetherian and formal, and let

$$f : \mathfrak{X} \longrightarrow S$$

be S-adic, essentially of finite type. Suppose $J^p(\mathfrak{X} \mid S)$ is proper over S. Then $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$ is coherent, and we define an ideal sheaf on S by

$$I^p_S(\mathfrak{X} \mid S) = I^0_S\big(f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big).$$

Let $J^p_S(\mathfrak{X} \mid S)$ be the closed formal subscheme of S defined by this ideal. If the fixed-dimension hypotheses of Theorem 5.4 are in force, one writes

$$I_S(\mathfrak{X} \mid S) = I^n_S(\mathfrak{X} \mid S), \qquad J_S(\mathfrak{X} \mid S) = J^n_S(\mathfrak{X} \mid S).$$

Corollary 5.10. Under the hypotheses of Definition 5.9:

(1) The underlying set of $J^p_S(\mathfrak{X} \mid S)$ is the image of $J^p(\mathfrak{X} \mid S)$ under $f : \mathfrak{X} \to S$.

(2) The formation of $J^p_S(\mathfrak{X} \mid S)$ commutes with base change.

(3) If

$$J^p(\mathfrak{X} \mid S) = \coprod_i J^p(\mathfrak{X} \mid S)_{(x_i)}$$

is the disjoint union of its isolated formal local components, then

$$I^p_S(\mathfrak{X} \mid S) = \prod_i I^p_{S,(x_i)}(\mathfrak{X} \mid S).$$

(4) In the case of completion of an ordinary noetherian S-scheme X, the formal closed subscheme $J^p_S(\widehat{X} \mid \widehat{S})$ is the completion of $J^p_S(X \mid S)$.

Proof. Assertion (1) is the usual support property of the zeroth Fitting ideal. Assertions (2) and (4) follow from base change for Fitting ideals. Assertion (3) follows from the product formula for I^0 applied to the decomposition of the coherent algebra $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$.

Remark 5.11. The Jacobian ideal above is classical: it commutes with base change and, under the mild dimension hypotheses of Theorem 5.4, gives the smoothness criterion. There are also more intrinsic ways of defining a canonical non-smooth locus in complete generality. For an A-algebra B, set

$$I_{B/A} = \bigcap_E \operatorname{Ann}_B D^1(B \mid A, E),$$

where E runs through the finite-type B-modules. This ideal commutes with localization by Corollary 3.13(c). Hence for a morphism $X \to Y$ one obtains an ideal sheaf $I_{X/Y}$. If $X \to Y$ is essentially of finite type, then f is smooth at x if and only if

$$x \notin \operatorname{Supp}(\mathcal{O}_X/I_{X/Y})$$

by 3.13(a). The same construction carries over to adic formal schemes essentially of finite type and is therefore also suited to deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k-scheme whose singularities are isolated non-degenerate quadratic singularities. In view of 4.13, under the hypotheses considered here the question becomes a local one.

We return to the notation of Section 4. Thus A is a fixed complete noetherian local ring with residue field k, and $\mathcal{C}_A$ is the category of local artinian A-algebras with residue field k. Let X be a k-scheme of finite type and let x be a closed point of X. Recall that x is a non-degenerate quadratic singularity, rational over k, if

$$\widehat{\mathcal{O}}_{X,x} \simeq k[[x_1, \dots, x_n]]/(f)$$

and

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = (x_1, \dots, x_n).$$

One may then arrange that f itself is a quadratic form.

Lemma 6.1. Let

$$f \in k[[x_1, \dots, x_n]]$$

and suppose

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = \mathfrak{m} = (x_1, \dots, x_n).$$

Then there are parameters $x_1', \dots, x_n'$, with

$$x_i \equiv x_i' \pmod{\mathfrak{m}^2},$$

such that

$$f = q(x_1', \dots, x_n')$$

for a quadratic form q.

Proof. Write $f = f_2 + h_1$, where f_2 is quadratic and $h_1 \in \mathfrak{m}^3$. Since the partial derivatives generate $\mathfrak{m}$, there are elements $e_i^{(1)} \in \mathfrak{m}^2$ such that

$$\sum_i e_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_1 \pmod{\mathfrak{m}^4}.$$

After replacing x_i by $x_i - e_i^{(1)}$, the remainder is pushed into $\mathfrak{m}^4$. Repeating this construction, one obtains $e_i^{(v)} \in \mathfrak{m}^{v+1}$ and parameters

$$x_i' = x_i - \sum_{v \geq 1} e_i^{(v)}$$

for which all terms of degree at least 3 disappear. Thus f becomes its quadratic part in the new variables.

Lemma 6.2. Let

$$A_0 \simeq k[[x_1, \ldots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then:

(1) The natural maps

$$J(\mathfrak{X} \mid R) \longrightarrow J_R(\mathfrak{X} \mid R) \longrightarrow \operatorname{Spec} A$$

are isomorphisms.

(2) The base Jacobian ideal is principal:

$$I_R(\mathfrak{X} \mid R) = (t), \qquad R = A[[t]],$$

where t is a free formal generator of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. By 4.10(b), the formal versal deformation has the explicit form

$$R = A[[t]], \qquad \mathfrak{X} = \operatorname{Spf}\bigl(R[[x_1, \ldots, x_n]]/(F - t)\bigr),$$

where F is obtained from f by lifting the coefficients to A. The exact sequence

$$A_0 \xrightarrow{dF} \bigoplus_i A_0 \, dx_i \longrightarrow \Omega^1_{A_0/R} \longrightarrow 0$$

shows that

$$I(\mathfrak{X} \mid R) = (\partial F/\partial x_1, \ldots, \partial F/\partial x_n) = (x_1, \ldots, x_n)$$

on the special fiber. Hence the Jacobian subscheme is obtained by setting $x_1 = \cdots = x_n = 0$, and the remaining equation is $t = 0$. Thus the maps in (1) are isomorphisms and $I_R(\mathfrak{X} \mid R) = (t)$. Finally

$$R[[x_1, \ldots, x_n]]/(F - t) \simeq A[[x_1, \ldots, x_n]]$$

as an A-algebra, proving formal smoothness over A.

Corollary 6.3. Let $x \in X_0$ be a non-degenerate quadratic singularity rational over k.

(1) Let $(R, \mathfrak{X}_1)$ and $(R, \mathfrak{X}_2)$ be two deformations of X_0 over an object R of $\mathcal{C}_A$, with the same point above x. Then the completed local R-adic formal schemes $(\mathfrak{X}_1)_{(x)}$ and $(\mathfrak{X}_2)_{(x)}$ are isomorphic over R if and only if

$$I_R((\mathfrak{X}_1)_{(x)} \mid R) = I_R((\mathfrak{X}_2)_{(x)} \mid R).$$

(2) Let R be a regular local ring and $(R, \mathfrak{X})$ a deformation of X_0. Then $\mathcal{O}_{\mathfrak{X},x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)} \mid R) \not\subset \mathfrak{m}_R^2,$$

that is, if and only if this Jacobian ideal defines a regular divisor in $\operatorname{Spf} R$.

Proof. Let the formal versal deformation be

$$A[[t]][[x_1, \dots, x_n]]/(F - t).$$

Two induced deformations over R are given by two elements $a, b \in R$:

$$R[[x_1, \dots, x_n]]/(F - a), \qquad R[[x_1, \dots, x_n]]/(F - b).$$

Their Jacobian ideals are (a) and (b). Equality of these ideals gives $a = vb$ for a unit v. Since R is complete and v has a square root after the standard lifting argument used here, the change of variables $x_i \mapsto v^{1/2} x_i$ identifies the two equations, because F is quadratic. The converse is immediate. The second assertion follows from the fact that the quotient of the regular local ring $R[[x_1, \dots, x_n]]$ by $F - a$ is regular precisely when $a \notin \mathfrak{m}_R^2$.

Theorem 6.4. Let X_0 be a k-scheme of finite type with only isolated non-smooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Let

$$\{z_1, \dots, z_m\}$$

be the set of non-smooth points of X_0. Suppose

$$H^2(X_0, D^0) = 0$$

and suppose that all z_i are non-degenerate quadratic singularities, rational over k. Then:

(1) R is formally smooth over A, i.e. R is a formal power-series ring over A.

(2) The Jacobian subscheme decomposes as a disjoint union

$$J(\mathfrak{X} \mid R) = \coprod_{i=1}^{m} J(\mathfrak{X}_{(z_i)} \mid R),$$

and for each i the natural maps from the local Jacobian component to its image in $\operatorname{Spf} R$ are isomorphisms.

(3) The ideal $I_R(\mathfrak{X}_{(z_i)} \mid R)$ is principal. If

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

then $t_1, \dots, t_m$ can be extended to a system of free formal generators of R over A, and

$$I_R(\mathfrak{X} \mid R) = \prod_{i=1}^{m} (t_i).$$

(4) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(5) If X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

Proof. For each singular point z_i, let $(R_i, \mathfrak{U}_i)$ be the formal versal deformation of $\operatorname{Spec}(\mathcal{O}_{X_0,z_i})$. Since $(R, \mathfrak{X})$ induces a deformation of this local scheme, there is a continuous local map $R_i \to R$ and a cartesian diagram

$$\begin{tikzcd} \mathfrak{X}_{(z_i)} \arrow[r] \arrow[d] & \mathfrak{U}_i \arrow[d] \\ \operatorname{Spf} R \arrow[r] & \operatorname{Spf} R_i. \end{tikzcd}$$

Lemma 6.2 gives

$$J(\mathfrak{X}_{(z_i)} \mid R) \simeq J(\mathfrak{U}_i \mid R_i) \times_{\operatorname{Spf} R_i} \operatorname{Spf} R$$

and

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

where t_i is the image in R of the corresponding local parameter. Since $H^2(X_0, D^0) = 0$, the local-to-global criterion of 4.13 implies that

$$\operatorname{Spf} R \longrightarrow \prod_i \operatorname{Spf} R_i$$

is formally smooth in the complementary directions; equivalently $t_1, \dots, t_m$ extend to a system of free formal generators of R. This proves (1), (2), and (3). Assertion (4) follows because each local singular piece is formally smooth over $\operatorname{Spec} A$ by Lemma 6.2(3), while away from the z_i the original scheme is already smooth.

For (5), formal smoothness of R over A, together with the exact sequence obtained in 4.4 and 4.11,

$$0 \to H^1(X_0, D^0) \to \mathfrak{m}_R/\mathfrak{m}_R^2 \to H^0(X_0, D^1) \to 0,$$

gives

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = \dim_k H^1(X_0, D^0) + \dim_k H^0(X_0, D^1).$$

Thus it remains to compute the Euler characteristic of D^0 on the curve. After extension of the base field to an algebraic closure, each non-degenerate quadratic singularity of a curve has completed local ring

$$k[[x, y]]/(xy).$$

If $A = \mathcal{O}_{X_0,z}$ and A' is its normalization, then

$$\ell(A'/A) = 1, \qquad \operatorname{Der}_k(A, \mathfrak{m}_A) = \operatorname{Der}_k(A, A),$$

and the conductor C of A in A' is $\mathfrak{m}_A$. Consequently

$$\operatorname{Der}_k(A, A) = \operatorname{Der}_k(A', A')$$

as A-submodules of $\operatorname{Der}_k(\operatorname{Frac} A, \operatorname{Frac} A)$, and one obtains an exact sequence

$$0 \to D^0_{X_0} \to D^0_{\widetilde{X}_0} \to \mathcal{O}_{\widetilde{X}_0}/C \to 0,$$

where $\widetilde{X}_0$ denotes the normalization. Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{\widetilde{X}_0}/C) = 2m.$$

Riemann–Roch on the normalization gives

$$\chi(D^0_{\widetilde{X}_0}) = 3 - 3(g - m).$$

Therefore

$$\chi(D^0_{X_0}) = \chi(D^0_{\widetilde{X}_0}) - \ell(\mathcal{O}_{\widetilde{X}_0}/C) = 3 - 3g + m,$$

which is equivalent to the stated formula.

Remark 6.5. In the formula of Theorem 6.4(5), the integer

$$a = \dim_k H^0(X_0, D^1)$$

is easily computed by Riemann–Roch. Namely

$$a = \begin{cases} 0, & g \geq 2, \\ 1, & g = 1, \\ 3, & g = 0. \end{cases}$$

Finally, in characteristic 2 there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension when $\operatorname{char} k = 2$. In that case one uses quadratic forms of maximum possible non-degeneracy.

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity, rational over k, if:

(i) either $\operatorname{char} k \neq 2$, or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd; in this case z is required to be a non-degenerate quadratic singularity rational over k;

(ii) or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is even; after extension to an algebraic closure $\bar{k}$, the completed local ring has the form

$$\bar{k}[[x_0, x_1, \ldots, x_{2n}]]/\big(x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}\big).$$

Lemma 6.7. Assume $\operatorname{char} k = 2$, and put

$$A_0 = k[[x_0, x_1, \ldots, x_{2n}]]/(f), \qquad f = x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then

$$R = A[[t_1, t_2]],$$

where t_1, t_2 are free formal generators over A,

$$I_R(\mathfrak{X} \mid R) = (t_1^2),$$

and $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. Since

$$\left(\frac{\partial f}{\partial x_0}, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_{2n}}\right) = (x_2, x_1, \ldots, x_{2n}, x_{2n-1}),$$

the deformation space has two formal parameters. By 4.14 the versal deformation may be written

$$R = A[[t_1, t_2]], \qquad \mathfrak{X} = \operatorname{Spf}\big(R[[x_0, \ldots, x_{2n}]]/(F)\big),$$

with

$$F = f + t_1 + t_2x_0.$$

The exact sequence for differentials gives

$$I(\mathfrak{X} \mid R) = (t_2, x_1, \ldots, x_{2n}),$$

and the residual equation on the Jacobian locus is t_1. Thus the induced base ideal is (t_1^2). Moreover the equation can be rewritten so that the total formal scheme is formally smooth over A, giving the last assertion.

Corollary 6.8. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Suppose

$$H^2(X_0, D^0) = 0,$$

$\operatorname{char} k = 2$, $\dim X_0$ is even, and all non-smooth points $z_1, \dots, z_m$ are ordinary quadratic singularities. Then:

(1) R is formally smooth over A.

(2) Each local base Jacobian ideal is principal of the form

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i^2),$$

and

$$I_R(\mathfrak{X} \mid R) = \prod_i (t_i^2).$$

Moreover $t_1, \dots, t_m$ can be completed to a system of free formal generators of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

The proof is the same as that of Theorem 6.4, using Lemma 6.7 in place of Lemma 6.2 for the local calculation in characteristic 2.